Democratic and Popular Algerian Republic
Ministry of Higher Education and Scientific Research
University of Science and Technology Houari Boumediene
Faculty of Computer Science
Department of Computer Science

License Thesis

Computer Sciense field

Computer Science Department

Speciality : Academic Computer Science (ACAD)

Theme :

Design and Development of a Platform for the Management and Monitoring of Energy Projects

Framed by:
M.ASMA Mohammed
Monitoring committee:
M.BOUCHAM Souhila

Directed by:
BOUMEZEBEUR Farah
BOUNOUA Aya
CHERIF Sara
KARABADJI Rania

Supported before the jury composed of:

M.BOUCHAM Souhila

M.KESSI Kahina

N°project 062/2024

Thanks

First of all, we thank God, the Almighty, for granting us the strength, courage, and willpower necessary to successfully complete this modest undertaking.

At the end of this project, we wish to express our deep gratitude to all those who, directly or indirectly, contributed to its completion.

We would particularly like to thank ***Mr. ASMA Mohamed****, Senior Engineer at* ***Sonelgaz – Electricity Production,*** *for his supervision, valuable advice, relevant guidance, and invaluable assistance throughout the writing of this thesis. We are grateful for his constructive comments, his constant availability, and the time he generously devoted to us. Thanks to him, we were able to enrich our knowledge and progress in the completion of this project.*

We also express our gratitude to all those who supported us throughout this journey.

We also thank ***Ms. Boucham*** *for serving as a* ***member of our monitoring committee*** *for this project.*

To my judges,

Ms. Boucham Souhila, *You have honored me by presiding over this jury.*

Ms. Kessi Kahina, *My sincere thanks for your participation on this jury.*

Dedications

We dedicate this work to:

Our respective parents, for their unconditional love, sacrifices, encouragement, and moral support throughout our journey.

Our families, for their patience, understanding, and caring presence during the most intense moments of this project.

Our loyal friends, for their support, encouragement, and motivating words when we needed them most. A special thought goes to Nejwa, Yousra, and Rachelle.

All the people who believed in us, near and far, and who inspired us to persever

Résumé

Les objectifs d’une entreprise de services sont d’une part la satisfaction de ses clients à travers des services répondant à leurs attentes et réalisés dans les budgets et les délais impartis, et d’autre part l’optimisation des ressources afin que l’entreprise puisse réaliser ses travaux de manière rentable.

Pour la filière Sonelgaz-PE, le recours à la mise en place de l'automatisation des processus est obligatoire car les projets ne peuvent être gérés par une personne dédiée à ce rôle. Cela implique un risque élevé de dérive s’il n’y a pas de système de gestion bien rôdé qui facilite la communication entre des différents acteurs.

De ce fait, L’objectif de notre projet est la conception et la réalisation d’une plate-forme pour la gestion de projet simple mais efficace. Visant à structurer, assurer et optimiser le bon déroulement d’un projet tout en facilitant la planification et le suivi d’un projet, ainsi d'assurer le bon déroulement des actions des employés, et suivre leurs missions dans les meilleures conditions.

Abstract

The objectives of a service company are on the one hand the satisfaction of its customers through services meeting their expectations and carried out within the budgets and deadlines, and on the other hand the optimization of resources so that the company can carry out its work profitably.

For the Sonelgaz-PE, recourse to the implementation of process automation is mandatory because the projects cannot be managed by a person dedicated to this role. This implies a high risk of drift if there is no well-functioning management system that facilitates communication between different actors.

Therefore, the objective of our project is the design and realization of a platform for simple but effective project management. Aiming to structure, ensure and optimize the smooth running of a project while facilitating the planning and monitoring of a project, thus ensuring the smooth running of employee actions, and monitoring their assignments under the best conditions.

ملخص

إن أهداف شركة الخدمات تتمثل من جهة في إرضاء عملائها من خلال تلبية الخدمات لتوقعاتهم وتنفيذها وفق الميزانيات والمواعيد النهائية، ومن جهة أخرى تحسين الموارد حتى تتمكن الشركة من تنفيذ أعمالها بشكل مربح.

بالنسبة لقطاع يعد اللجوء الى تطوير سير العمليات وحوسبتها أمرًا إلزاميًا.. إنه لا يمكن إدارة المشاريع في سونلغاز من قبل شخص مكرس لهذا الدور. وهذا يعني وجود مخاطر عالية من الانجراف إذا لم يكن هناك نظام إدارة يعمل بشكل جيد يسهل التواصل بين مختلف الجهات الفاعلة.

لذلك، فإن الهدف من مشروعنا هو تصميم وتنفيذ منصة بسيطة وفعالة لإدارة المشروع. تهدف إلى تنظيم وضمان وتحسين التشغيل السلس للمشروع مع تسهيل تخطيط ومراقبة المشروع، وبالتالي ضمان التشغيل السلس لإجراءات الموظفين، ومراقبة مهامهم تحت ظل أفضل الظروف.

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General Introduction:

Currently, computing manages various fields in a precise and efficient manner. So, companies are subjected to the necessity of automating their work processes in project execution in order to achieve better results.
The execution of a project encompasses all the actions carried out by a team. It involves financial, personnel, and material resources. To achieve the set objective, good management is essential as it makes the work process well-organized and more efficient.

Indeed, a successful project is above all a project managed and conducted effectively. It is no longer enough to manage a project methodically; it is necessary to implement genuine project management strategies to mobilize the strength and energy of all the company's stakeholders, at all hierarchical levels, towards achieving objectives. And thus develop a new corporate culture.
This work aims to set up "a web application for the management and monitoring of energy projects," and within this framework, our graduation project is included. Its objective is to plan projects and provide good tracking of project actions. And provide high-quality, precise real-time information to facilitate sharing and communication among project stakeholders, ensure and optimize the smooth progress of a sufficiently complex project, and help the project manager break it down into manageable sub-units, which is essential for project execution. And thus to its successful completion and success. Thus, to assign tasks and monitor the progress of the project.

Our system must manage the following elements in particular:
• **Account management:** Allows the administrator to add, modify, view, or delete accounts.
• **Task management:** Allows adding, modifying, viewing, deleting, or assigning tasks.
• Project tracking: Allows calculating the remaining work and the progress level for a task as well as for a project.

<u>Thesis outline:</u>

This report aims to present the work carried out during our four-month internship. To conduct our study properly, we have divided the document into three chapters organized as follows:

• The 1st chapter: titled "Generalities and State of the Art," describes the context of the project, namely the internship location, the project description, and the proposed solutions.

• The 2nd chapter: titled "Needs Analysis and Design," which focuses on the company's needs as well as UML modeling of the proposed solution.

• The 3rd chapter: titled "Implementation," This chapter is dedicated to the implementation of the application, providing a detailed presentation of the developed software as well as an overview of the various tools used in this stage and the different graphical interfaces of the application.

Finally, we concluded the aforementioned final report with a conclusion and perspectives for potential improvements.

Problem and motivation:

Managing projects manually is a costly approach in several ways. The current procedure of exchanging data within Sonelgaz can encounter many problems, leading to a considerable loss of time, resources, efforts, and other issues that do not meet the company's objectives.

• **Time loss:** exchanging documents between project members and calculating the percentage of project progress takes a lot of time to accomplish the desired work compared to using software, which significantly reduces the risk of delays.

• **Loss of information:** The burden of documents that can be lost, damaged, or stolen would be better protected if they were stored on a server. Loss of information: The burden of documents that can be lost, damaged, or stolen will be better protected if they are stored on a server.

• **Non-shared documents:** Two employees cannot work on the same document at the same time; however, using a developed IT solution allows for bidirectional information exchange.

Absence of a dashboard that keeps the concerned parties updated in a project, making it difficult to detect delays and overruns in the project timeline. It is necessary to implement a management system that ensures proper tracking of project actions to send alerts and warnings to project members and inform them of its status.

It's true that there are already standard software for project management (Microsoft Project, Wrike, Atlassian JIRA, etc.), but some do not meet the company's needs because they have limited functionalities: Some are very cumbersome, non-collaborative, expensive, and costly to set up and maintain, while others are very difficult to use and therefore require employee training.

Consequently, the improvement and development of information and communication technologies is deemed necessary. The best solution is to design a web application that manages all these risks and meets the company's needs to enable the evolution of project management, so everything will revolve around a number of methods and best practices.

Chapter 1 :

Generalities and state of the art

1. Introduction:

In this chapter, we will first present the host organization that opened its doors to welcome us within its IT development department, followed by a study of the existing situation and a critique allowing the project to present an improvement summarizing all the chosen solutions.

2. Presentation of Sonelgaz-PE:

The company Sonelgaz-Production of Electricity (S-PE) is a key and historical player in the national electricity production scene, possessing the largest production capacity park of over 18 GW of developable power to date, which gives it the position of the leading operator on the interconnected network.
S-PE is present throughout the national territory with its Regional Production Departments and their attached units, in order to meet the ever-increasing demand for electrical energy by utilizing and optimizing its primary resources while preserving the environment by producing environmentally friendly and cost-effective energy.
S-PE is a joint-stock company with a capital of 35 billion Dinars, whose headquarters is located at National Road No. 38, 600 Offices Building – Gué de Constantine – Algiers.

S-PE Organizational Chart ::

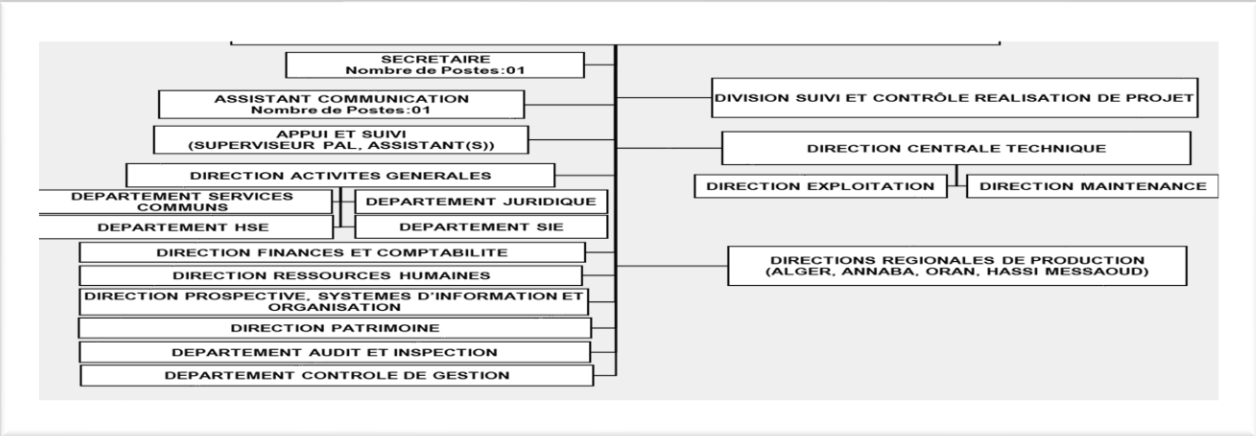


Figure 1:Organizational Chart of the General Management S-PE

more details go to **appendix1**

3. Existing solutions on the market for project management:

The stakes of project management are very high, especially for a specialized company looking to ensure the success of a project in order to improve its processes to identify and meet the needs of its clients. To conduct our research, we were assisted by the websites of software publishers and also by blogs that provided us with genuine reviews of the various products.
Therefore, we distinguished two types of products:

3.1 Proprietary solutions:

When it comes to managing a project from A to Z with multiple stakeholders and deadlines, using project management software often proves indispensable, especially in 2025 when most collaborators are dispersed.

For that, there are many tools available.
At the forefront of the market is the Microsoft Project application, which allows for project and resource planning, as well as monitoring projects during their execution. MS Project allows the project manager to ensure professional project management, in accordance with the state of the art, and thus guarantee adherence to deadlines and budget. There are still other software options like (Prevavira, MsProject, WRIKE, JIRA, BaseCamp, Trello).

3.2 Open source solutions:

Open Source players are starting to emerge and carve out a place for themselves in the market. If they are still few in number, some offer features as complete as proprietary solutions. There are a large number of project management tools available in the open-source domain such as Dotproject, Project-Open, ProjectPro…

4. The limitations of existing market solutions for project management:

Many of these software solutions rely on methods that are often essential but have shown certain limitations:
Real project management is very difficult: collecting progress information is laborious and unreliable (information loss).

- Little or no means to anticipate potential problems .
- Communication between project stakeholders remains very difficult and costly.
- Few people plan the work of others, control is done by hierarchy, responsibilities are defined by the organization's structure rather than the project's, and communication is organized around meetings and emails.
- Complicated and time-consuming onboarding for newcomers. Little suited for internal communication.
- Access rights management is reserved for the most expensive version.

5. Definition of Project management:

Project management involves organizing the execution of a project from A to Z, from its design phase to its final phase. To do this, it is necessary to define the objectives, the budget, the deadlines, and any potential constraints.

6. The main stages of project management:

Managing projects is not something you can improvise. Each project being totally different from another, there is no miracle method to succeed in a project. It is advisable to select the most appropriate method for each project. Some major steps, however, generally carry over from one project management technique to another:

- **Design:** we determine the objectives to be achieved while considering the constraints that could hinder the smooth progress (human resources, timeline, tools, budget, technology…).
- **Planning:** Is the project feasible? It's time to plan it and determine its timeline (which milestones need to be achieved and when?). We flesh out the project at this stage: each step is described, and everyone's roles are determined. Human resources are tasked if the necessary skills are not present among the company's actors (search for new employees, speakers, intervention of freelancers...).
- **Execution:** once the roles and resources are defined, the previously set action plan is executed.
- **Control:** The manager works closely with their team to ensure that the planned schedule is being followed. To do this, he can, for example, organize weekly meetings to take stock.
- **Closure:** we fine-tune the final details and wrap up the project.

- **Review:** the teams gather to assess the success of the project. What worked? What difficulties were encountered? What could be improved in the future?

7. **Proposed solution:**

Due to the limitations mentioned in the previous paragraph, we have concluded that the perfect project management platform does not yet exist, either because each user in each company has a different management style for their tasks, or because the functionalities rarely combine ease, precision, and ergonomics, which is certainly complex. We must then establish a set of unified procedures for each project with the following objectives:

- Provide an efficient method to create a project action plan that promotes collaboration and allows managers to easily consult and update their actions themselves:
  - The project manager can create the different phases of the project with their specific activities, plan them, and assign them to the collaborators.
  - Collaborators can view the project's status at any time through a graph displayed on the dashboard and thus calculate the remaining tasks.
  - The administrator is supposed to manage the accounts.
  - The project manager has access to project reports and their progress status.
- Eliminate low-value-added tasks and save time (communications, follow-ups, reminders, preparation and dissemination of results).
- Improve and simplify tracking with the dashboard that allows for quick and effortless visualization of project progress, action tracking, and identification of all incidents.
- Improve and simplify tracking with the dashboard that allows for quick and effortless visualization of project progress, action tracking, and identification of all incidents.
- Centralize data and documents into a single, integrated, modern, and more efficient system.
- Improve access to and sharing of information
- Create accounts on the platform for each project member to gather, their tasks, and all project information in one place. This allows for maintaining control over the personnel and managing their access to the platform.

In addition: (non-functional objectives)

- Develop a multi-user platform accessible from any workstation connected to the company's server
- Rich, intuitive, and fast-to-use interface

8. **Conclusion:**

We concluded that our platform plays a good role in project management. It will therefore analyze various project management issues while ensuring the best project tracking and a dynamic management of activity (task) progress. To achieve this, we have intensified our efforts to foster a stronger and more rigorous project management culture.

Chapter 2 :

Requirements Analysis and Design

1. Introduction :

Design and implementation of a platform for the management and monitoring of energy projects After presenting in the previous chapter the host organization, the existing solutions on the market and their limitations, and the proposed solution. We will begin the second chapter with the analysis of needs, defining the various functional and non-functional requirements of our system. Then, we will identify the actors and their roles, as well as the different use cases of our system. We will conclude with the modeling and design of our system.
In this chapter, we will present a very important phase of a project's life cycle that plays the role of structuring, planning, and organizing a project.
We will use the UML language for modeling in the analysis and design of this project, and we will use three types of diagrams (the use case diagram, class diagram, and sequence diagram).
UML is a general-purpose visual modeling language used to specify, visualize, construct, and document the artifacts of a software system. It captures the decisions and understanding of the systems that need to be built. (Grady Booch, 2005)

2. Needs analysis :

Before starting any project, it is important to go through the needs analysis; it is the first step in the system development cycle. It should translate what the future system is likely to bring to users, disregarding how it will be built. She defines the system's functionalities and especially how to use it. This first phase, therefore, focuses on the external properties of the system, aiming to answer the question "what does the system do?", that is:
What the system can provide to the user.
The needs analysis phase involves understanding the client's needs and requirements, providing criteria and specifications that enable the design of a good solution. This will be done by identifying the actors and defining all the needs, which will then be modeled by a use case diagram, textual descriptions followed by their sequence diagrams, and finally the class diagram.

3. Specification of requirements:

In this section, we identify all the functionalities of our future application for each type of user by listing the functional requirements that lead to the development of use cases, and which allow us to establish the list of system requirements for its realization and proper functioning, translated into technical (non-functional) needs.
In this section, we identify all the functionalities of our future application for each type of user by listing the functional requirements that lead to the development of use cases, and which allow us to establish the list of system requirements for its realization and proper functioning, translated into technical (non-functional) needs.

a. Functional requirements:

- **Account management:** Add, View, Edit, and Delete an account.
- **Project management:** Add, View, Edit, and Delete a project.
- **Management of project Tasks**: Add, View, Edit, and Delete task.
- **Employee management**: Add, View, Edit, Assign, and Delete an employee.
- **Management of project Tasks monitoring:** Add, View, Edit, and Delete Task monitoring
- **Invoice management**: Add, View, Edit, and Delete invoices.
- **Document management:** Add, View, Edit, and Delete a document.
- **Incident management:** Add, View, Edit, and Delete an incident.
- **Management of project owner(customer):** Add, View, Edit, and Delete a project owner.
- **Management of service provider:** Add, Consult, Modify, and Delete a project manager.
- **Market management:** Add, Consult, Modify, and Delete a market.
- **Meeting management:** Add, View, Edit, and Delete a meeting.
- **Project monitoring:** Calculate the progress level of a task and a project.

- **View the dashboard:**
  - View the project progress rate graph .
  - Consult the incidents .
  - Consult the consumption rate of the AP.

b. **<u>Non-functional requirements:</u>**

The non-functional, or technical, requirements of our application can be summarized as:

- **Security and confidentiality:** The application must ensure the security of its users and confidentiality.
- **Ergonomics and flexibility:** The application must offer a user-friendly interface, easy to use and exploitable by the user.
- **Availability:** The application must be available at all times.
- **Speed:** The application must optimize processes to respond quickly to user actions.
- **Maintainability:** The application must be extensible, and its code must be readable to ensure its evolutionary state.

4. **<u>Identification of stakeholders:</u>**

Here are the different actors who interact with our system:

- **Administrator:** this is a key actor, their role is to manage the various accounts and user privileges.
- **Project manager:** he is a key actor, his role is to organize and lead the project until its deployment, managing the different tasks of the project as well as any incidents that may arise.
- **Employees:** they are a key player, the members who work on the project, each employee is responsible for completing their respective tasks.
- **Manager:** responsible for monitoring projects through dashboards and tracking incidents and financial dashboards.
- **Financial team:** they are a key player, the members who work on the financial monitoring of the project.

5. Use case diagrams:
5.1. Global use case diagram :

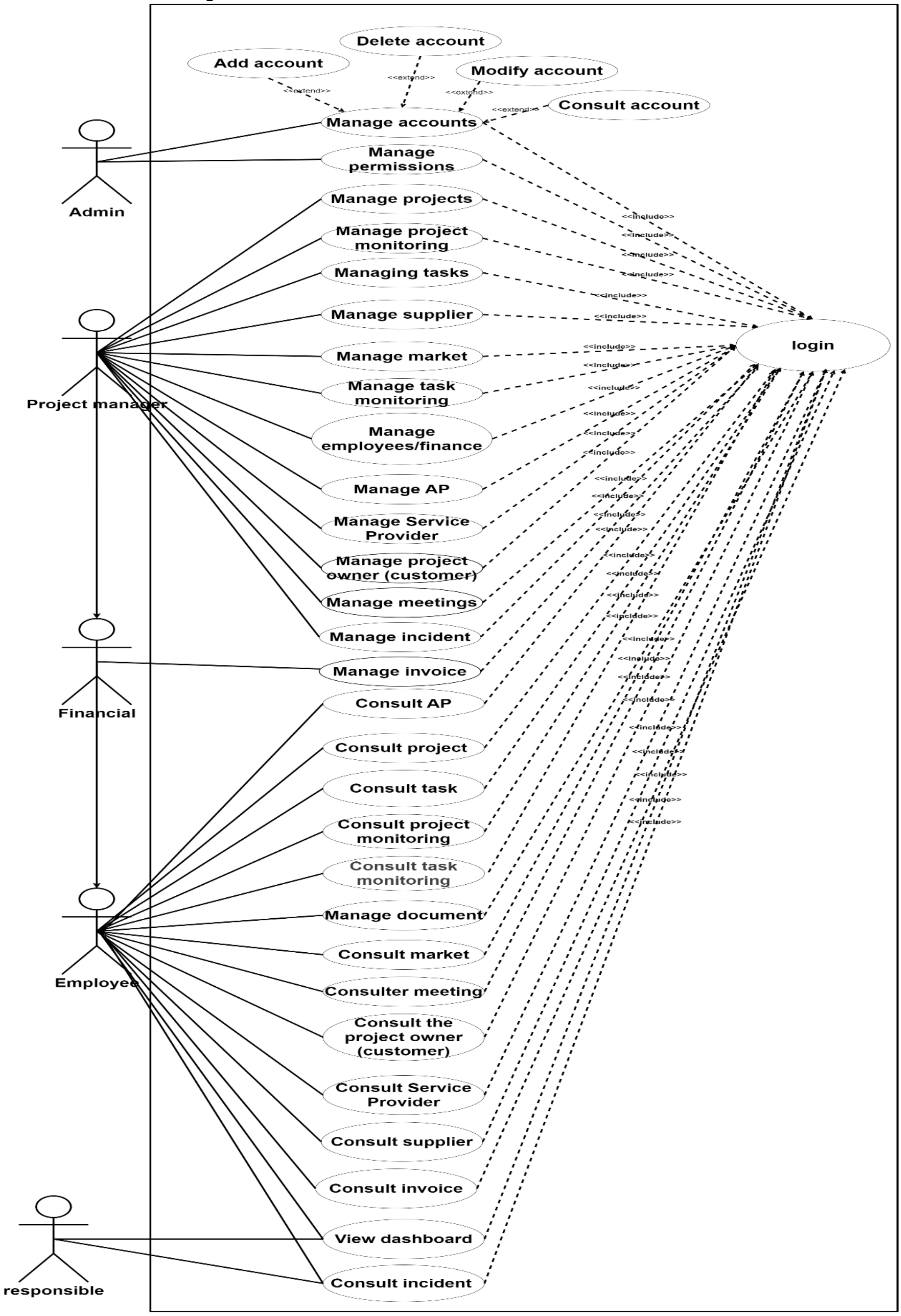


Figure 2:Global use case diagram

5.2. Project manager use case diagram:

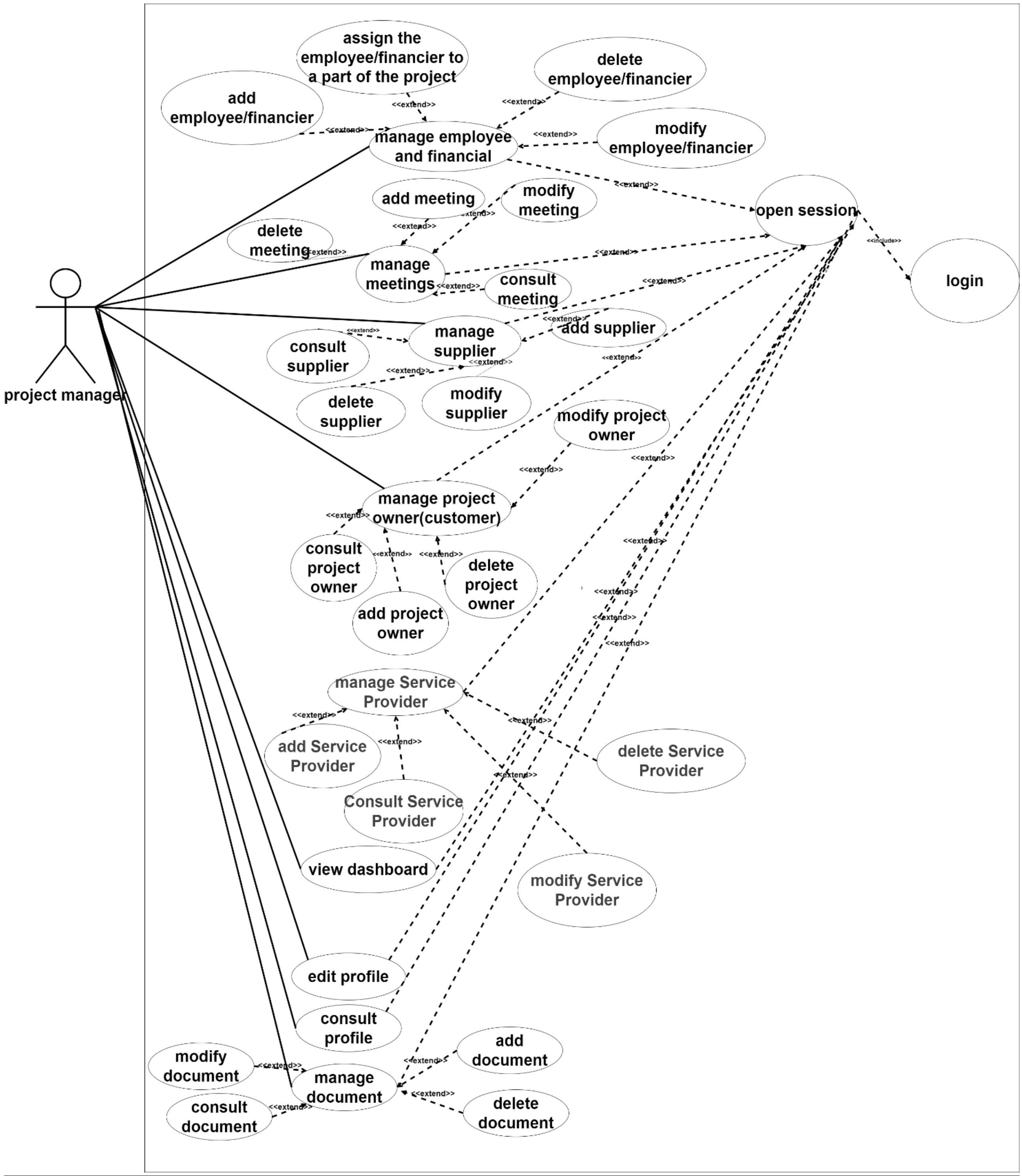


Figure 3:Project manager use case diagram part1

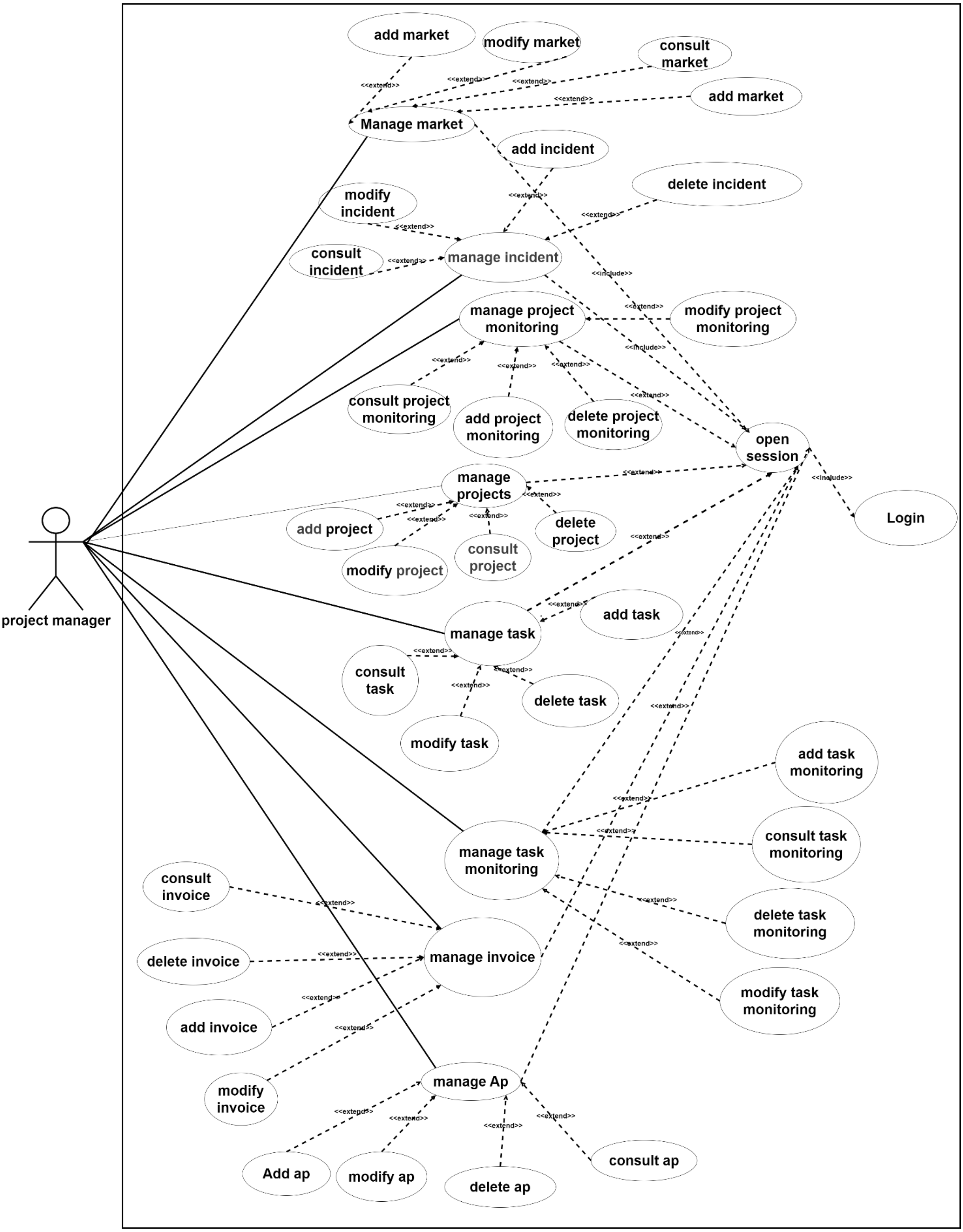


Figure 4:Project manager use case diagram part2

5.3. Employee and financier use case diagram:

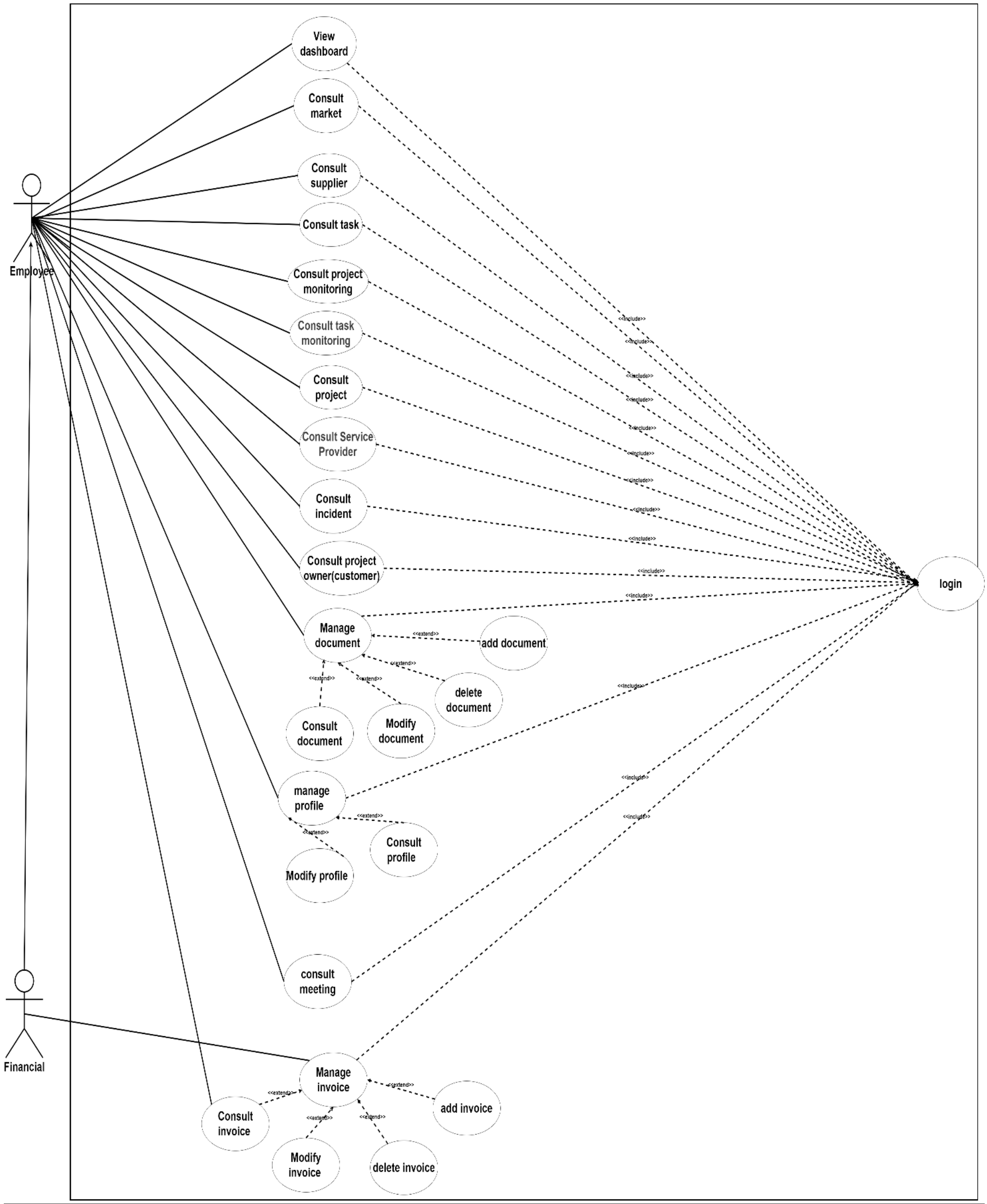


Figure 5:Employee and financier use case diagram

6. The analysis and static design of the application:

Class diagram:

A class diagram is a static representation of a system in terms of classes and their relationships. A class is an abstract concept that represents its attributes and behavior.

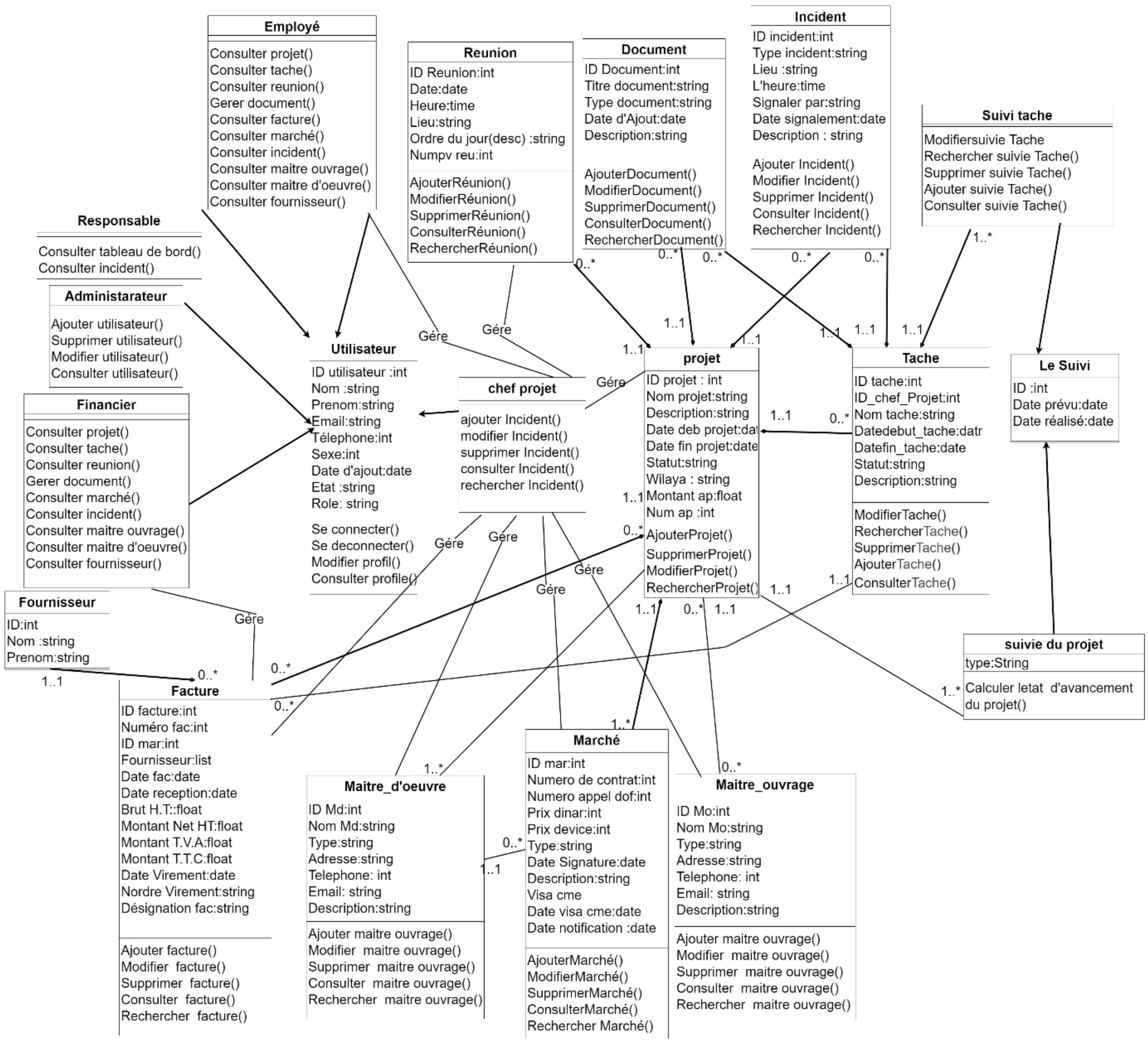


Figure 6 : Class diagram

For the definitions of class descriptions, go to Appendix2

7. Sequence diagram of project management:

Scenario: The user can manage projects (add, modify, delete, and extract reports) after gaining access.

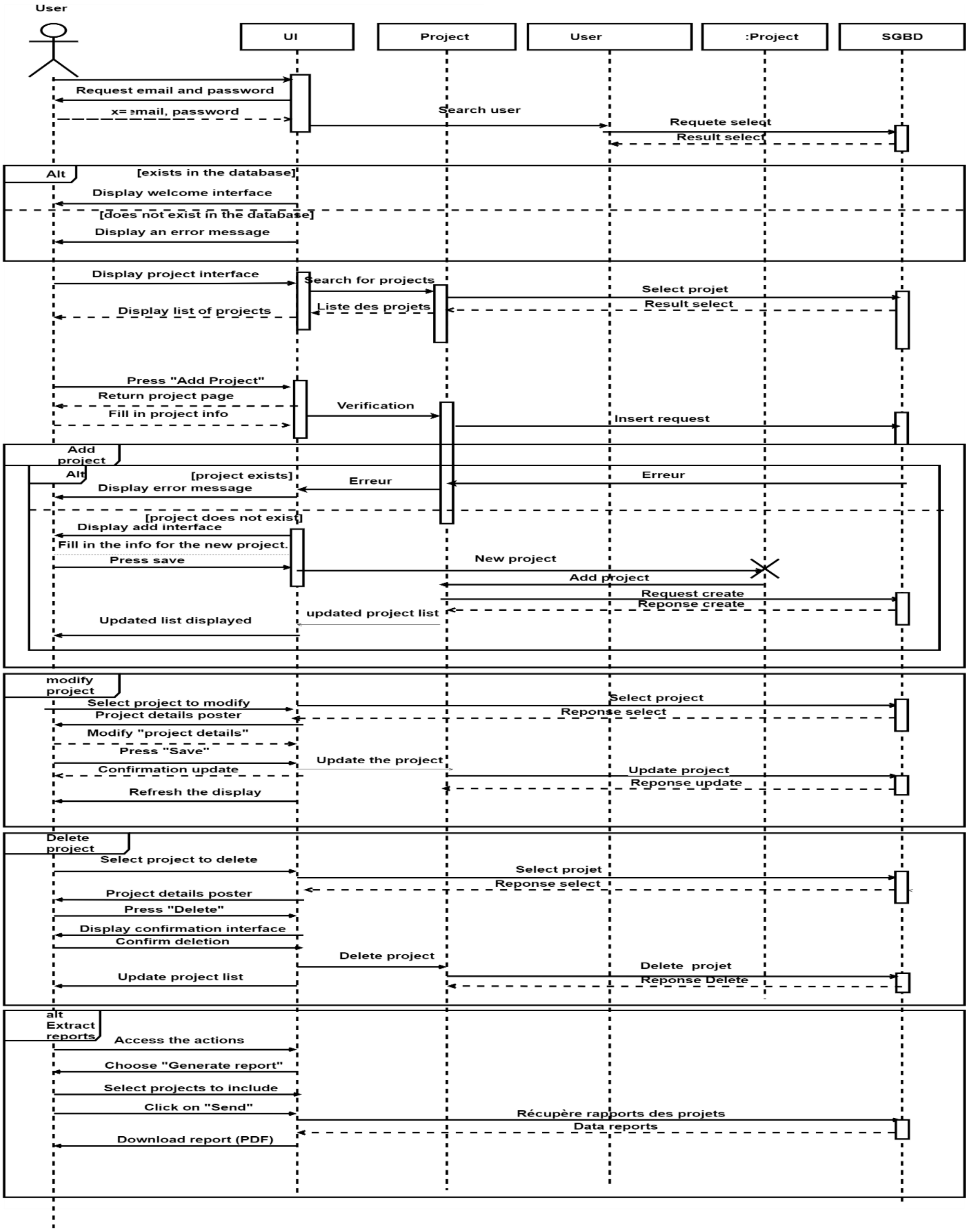


Figure 7 : Sequence diagram

8. Conclusion:

In this chapter, we presented a preliminary study called "analysis and specification of project needs" and proposed a design and modeling of the application to be developed.

We started with a needs analysis followed by the modeling of the solution using UML diagrams: use case diagrams, sequence diagrams, as well as class diagrams, with the use of design aid tools for more rigor and attention to successfully complete our project, which will subsequently allow us to develop an efficient and high-performing application.

In the next phase, we are now able to begin the implementation and development of our application.

Chapter 3 :

Realization

1. Introduction :

After finalizing the analysis and design of the previous chapter, we will, in this chapter 3, begin the implementation and realization part of the solution. We will start with the software environment by showing the development and design tools useful in our project along with the various programming languages, and then we will proceed to the presentation of the different interfaces implemented using screenshots illustrating the progress of our solution.

2. The Tools Used:

In this section, we will present the design and development tools used to create our application.

2.1. Design tools:

Draw.io :

Figure 8 : logo draw.io

draw.io is a technology stack for creating diagram applications and the most widely used browser-based end-user diagramming software in the world. (Ltd, 2025)

2.2. Development tools:

MySQL :

Figure 9:logo MySQL

MySQL is an open-source relational database management system (RDBMS) used to store and manage data. Its reliability, performance, scalability, and ease of use make MySQL a popular choice among developers. You will also find it at the heart of demanding and high-traffic applications such as Facebook, Netflix, Uber, Airbnb, Shopify, and Booking.com. (Oracle, 2025).

Vs code :

Figure 10:logo Vs code

Visual Studio Code, commonly known as VS Code, is a free and lightweight code editor that has quickly become one of the most popular tools for developers of all skill levels across a range of programming languages. (Microsoft Corporation, 2025)

3. Implementation languages:

Django :

Figure 11:logo django

Used for the back-end. Django is an open-source Python framework. It is therefore clearly oriented towards developers who need to produce a solid project quickly and without surprises. Django offers a solid project foundation. Django is therefore a great toolbox that helps and guides the developer in building their projects.(Django Software Foundation, 2025)

React.js:

Figure 12:logo React

Used for the front-end. React is a JavaScript library. Thanks to React, it is easy to create interactive user interfaces. Define simple views for each state of your application, and when your data changes, React will optimally update only the components that need it. (Meta Platforms, 2025)

4. System interfaces:

4.1. Authentication Interface:

As mentioned in the previous chapter, our platform is divided into several spaces dedicated to different types of users. An interface is reserved for the project manager, which allows them to organize and lead the project until its deployment. Another is intended for employees, who are the members working on the project; each employee is responsible for completing their assigned tasks. A dashboard is provided for the manager, who is responsible for tracking projects and incidents, budget monitoring, and project-related expenses. An interface is also dedicated to the finance team, who are responsible for budget tracking and project-related expenses. Finally, the administrator has full access to manage user accounts and the privileges of each actor.
When any user of the platform launches their application, the appropriate authentication page is displayed. He must provide his email and password to be able to log into the platform.

Figure 13:Authentification interface

4.2. Interface admin :

The admin is the only user who has the right to add a user. the first user account on the platform will be created by us and assigned to the first administrator. Once authenticated as an administrator, they will have a list of all users, allowing them to manage all accounts (add, modify, view, and delete). This centralized management ensures secure and structured control of access to the platform.

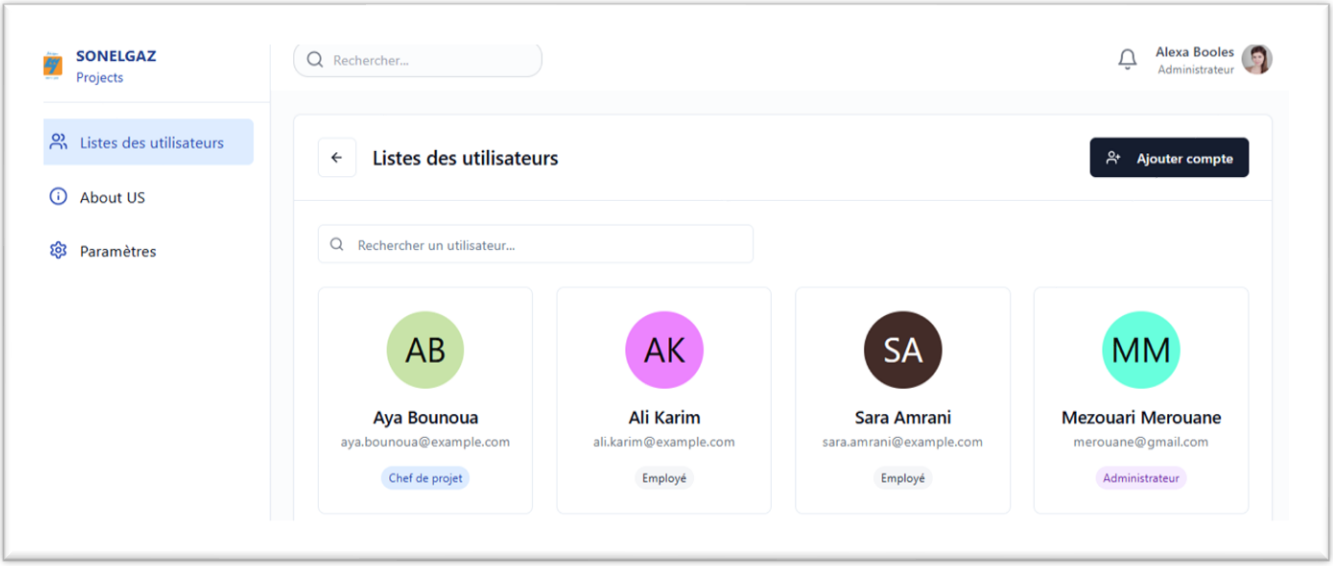


Figure 14:Admin interface

After creating the account, the system automatically sends a confirmation link to the provided email address. Thanks to this link, the user can set their password and then proceed to authenticate themselves on the platform. this technique will really strengthen the platform's security, as even the administrator himself does not have access to users' passwords, thus ensuring the confidentiality and protection of personal data.
To search for a user, simply enter their name.

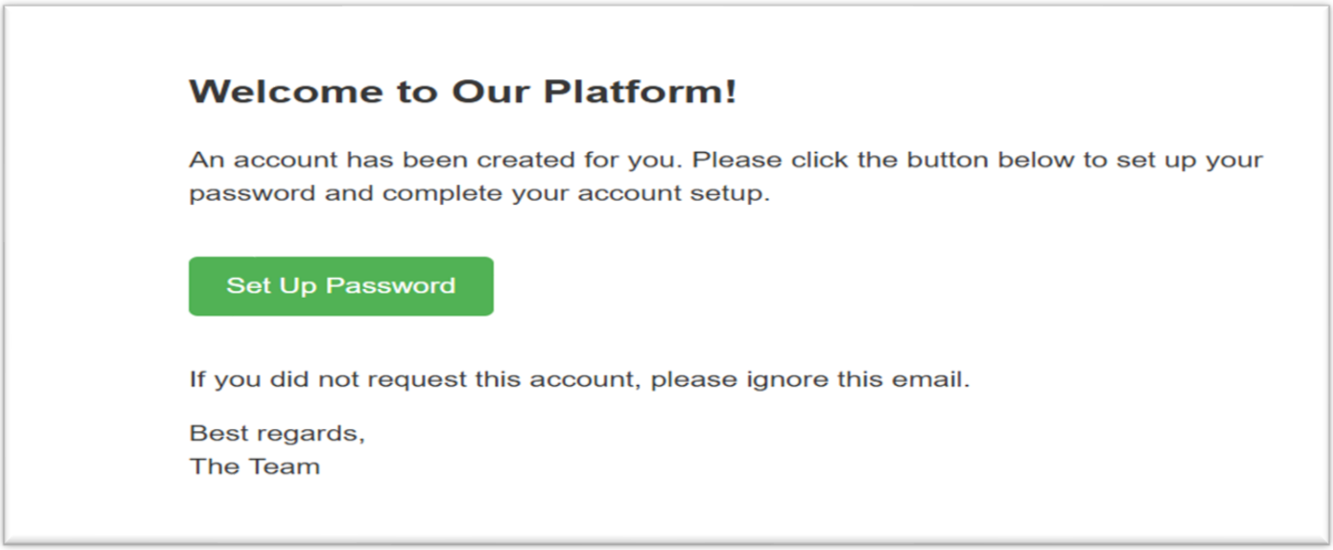


Figure 15:Password verification

4.3. Interface project :

Once logged in, the user is redirected to a personalized interface, displaying only the projects they are associated with.
The "project manager" user has extended rights: they can add, modify, delete, and view projects, as well as search for a project by its name.
The "employee" user has read-only access: they can view and search the projects they are assigned to, but cannot modify or delete them.
As for the "financial" user, they can consult, filter, and search for projects, particularly based on financial criteria, in order to ensure effective monitoring of budgetary aspects.

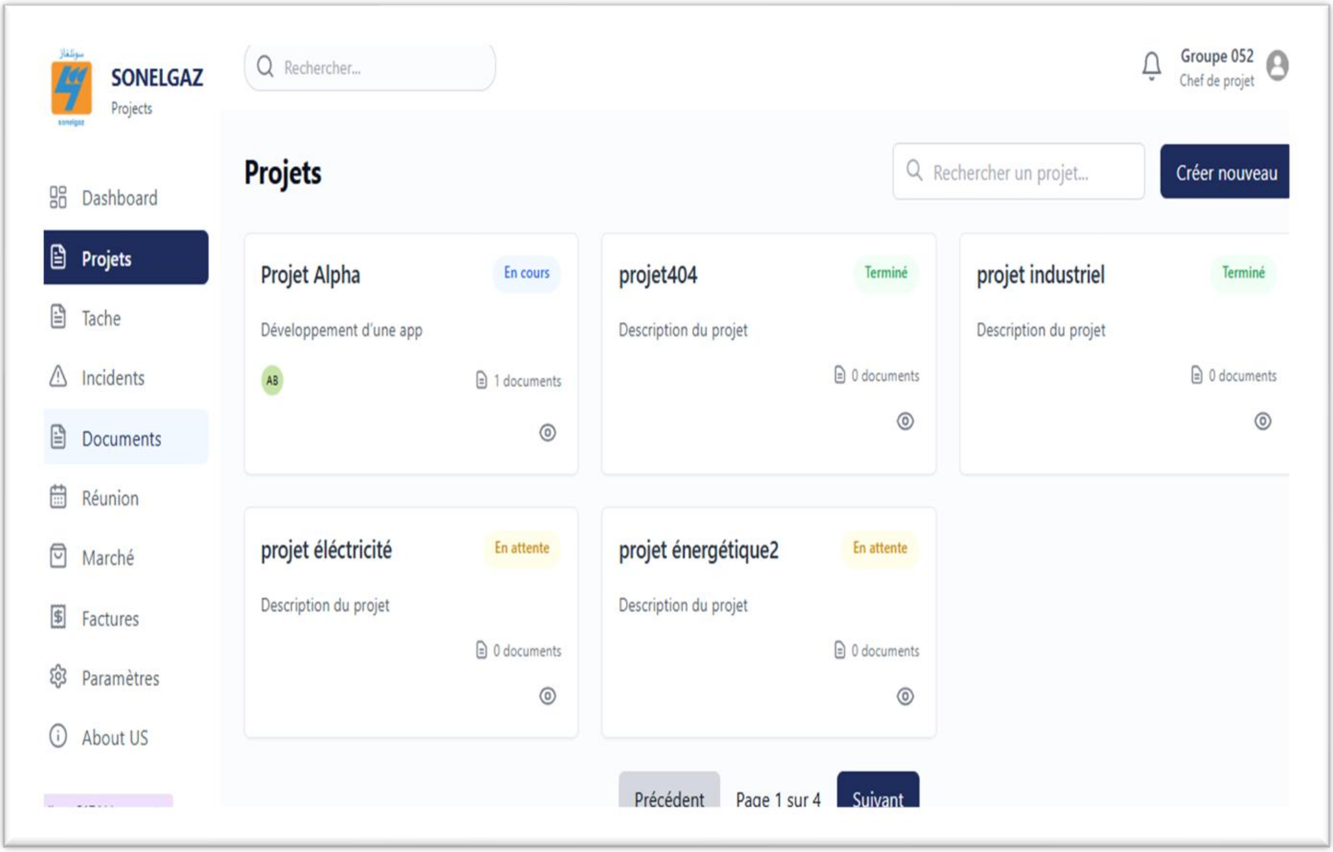


Figure 16:Projects

Dashboard:

The dashboard includes:
Project progress represented by a timeline.
Budget consumption represented by a trigonometric circle.
The list of Incidents.

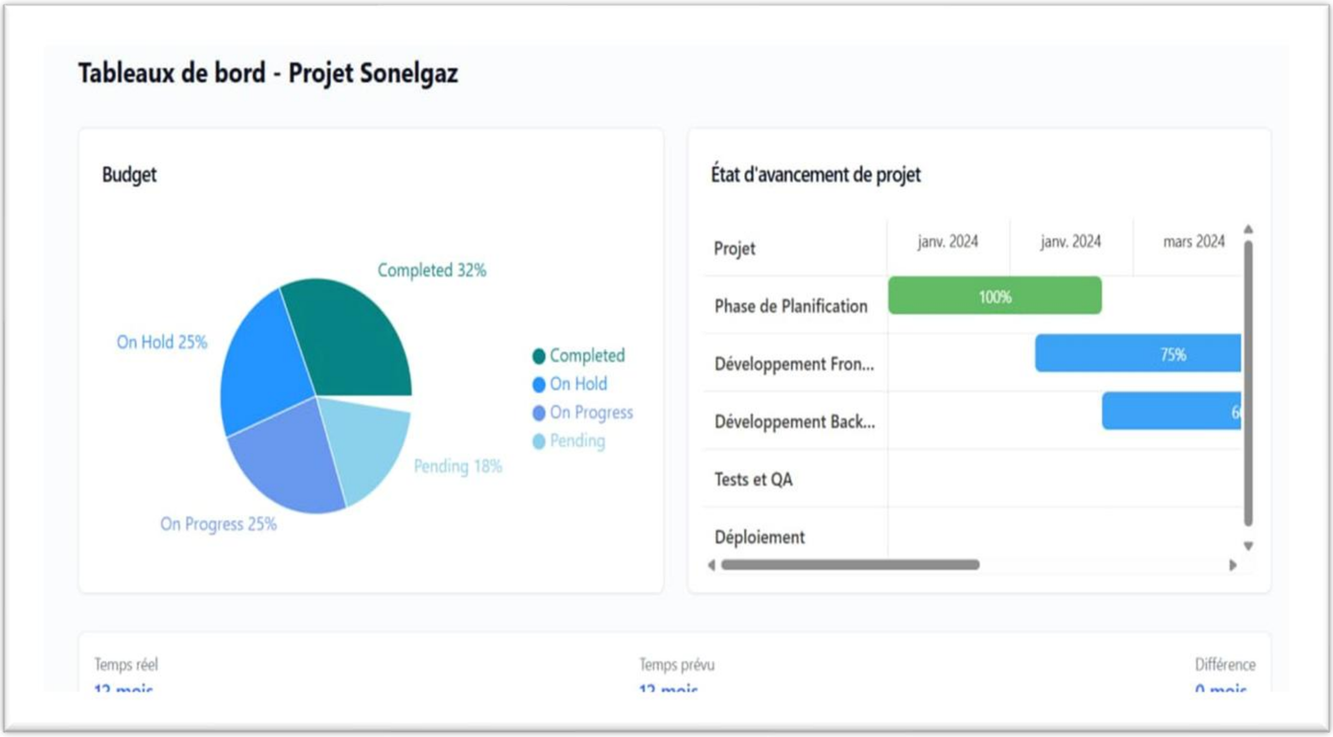
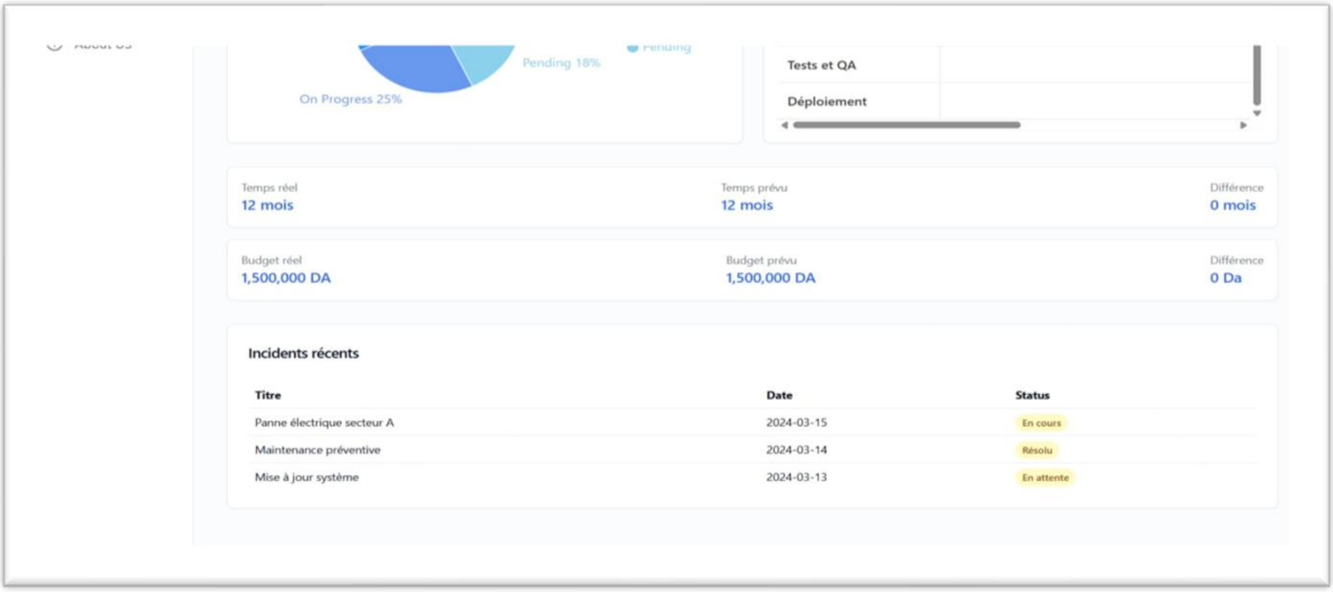


Figure 17:Dashboard



For more details, go to Appendix3

Conclusion:

We presented in this last chapter the software environment in which we developed our application. Subsequently, we presented some graphical interfaces of our application to illustrate their various functionalities using screenshots.

General conclusion:

The developed system was created to meet the demand of the Sonelgaz-PE/SPA organization within the Sonelgaz Group, which aimed to automate the process of monitoring the implementation of energy projects and to quickly exchange data and information. The work entrusted to us allowed us to meet the company's needs and ensure effective project management, providing quality information.

To design the system, we went through the traditional development stages, which include the requirements definition phase, the design, and eventually an agile implementation of the system.

This document summarizes all the steps we went through to achieve the set goals and objectives, which began with recognizing the host organization Sonelgaz-PE and its internal organization, then studying the problem and the limitations of existing solutions in the market in order to propose our solution.

Subsequently, we analyzed the specifications of functional and non-functional requirements, which allowed us to identify the various actors interacting with the targeted application, and then modeled all these identified functionalities based on UML modeling (use case diagram, sequence diagram, and class diagram). The objective of the next part was the detailed design, in which we established the overall structure of the application.

The last part of our project was the application implementation phase, which was dedicated to presenting the tools used for the development and deployment of the solution, as well as the most significant interfaces of our information system.

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Appendix 1

Company presentation :

Missions and responsibilities of the General Directorate S-PE:

The company that operates in the electricity production and marketing segment has as its main mission the operation and maintenance of its power plants and related equipment, or those of third parties, whether in Algeria or abroad.

In this capacity, it fulfills the following missions and obligations:

- The development of strategies and doctrine in the technical and management fields in collaboration with the structures of the Sonelgaz Group;
- The operation, maintenance, and safety of associated installations and equipment;
- Compliance with environmental protection regulations;
- Compliance with public service obligations regarding the regularity and quality of electricity supply as well as customer provisioning;
- Respect for public service obligations regarding the regularity and quality of electricity supply as well as customer provisioning;
- The Company also aims to directly or indirectly participate in all industrial, commercial, or financial activities or operations, whether movable or immovable, in Algeria or abroad, in any form whatsoever, as long as these activities or operations can be directly or indirectly related to its corporate purpose or to any similar, related, or complementary purposes; The Company also aims to take direct or indirect participation in all industrial, commercial, or financial activities or operations, movable or immovable, in Algeria or abroad, in any form whatsoever, as long as these activities or operations can be directly or indirectly related to its corporate purpose or to any similar, related, or complementary objects;
- The Company may achieve its objective indirectly, notably through the leasing of businesses, contributions to any company to be created or existing, or joint ventures, by taking interests or participating in any companies or enterprises through subscriptions, purchases of shares, or social rights.

Appendix 2

Description of class:

Classe	Attribut	Description	Remarque
Projet	ID projet	Identifier	Auto Incrément
	Nom projet	Name of the project	
	Description	Description of the projet	
	Date deb projet	The strat date	
	Date fin projet	The end date	
	Statut	Project status	
	wilaya	Wilaya of the project	
	Montant AP	The project finance(budget)	
	Num ap	Number of program autorisation	
Utilisateur	ID utilisateur	Identifier	Auto Incrément
	Nom	User’s name	
	Email	User’s email	
	Prenom	User’s first name	
	Role	Role of the user	
	Sexe	Gender of the user	
	Etat	State of the user	
	Date d’ajout	Date of addition of the user	
	Télephone	User’s phone number	
Administrateur	ID Administrateur	Identifier	Auto Incrément
	Nom	administrator’s name	
	Email	administrator’s email	
	Prenom	administrator’s first name	
	Role	Role of the administrator	
	Sexe	Gender of the administrator	
	Etat	State of the administrator	
	Date d’ajout	Date of addition of the administrator	
	Télephone	administrator’s phone number	
Chef de projet	ID chef de projet	Identifier	

	Nom	Project manager’s name	
	Email	Project manager’s email	
	Prenom	Project manager’s first name	
	Role	Role of the Project manager	
	Sexe	Gender of the Project manager	
	Etat	State of the Project manager	
	Date d’ajout	Date of addition of the Project manager	
	Télephone	Project manager’s phone number	
Responsable	ID responsable	Identifier	
	Nom	Responsible’s name	
	Email	Responsible’s email	
	Prenom	Responsible’s first name	
	Role	Role of the Responsible	
	Sexe	Gender of the Responsible	
	Etat	State of the Responsible	
	Date d’ajout	Date of addition of the Responsible	
	Télephone	Responsible’s phone number	
document	ID document	Identifier	Auto Incrément
	ID tache	Task’s identifier	
	Titre document	Name of the document	
	Type document	Type of the document	
	Date d’ajout	Date of addition of the document	
	Description	Description of the document	
	Nom projet	Name of the associated project	
	Nom tache	Name of the associated task	
Employé	ID employé	Identifier	
	Nom	Employee’s name	
	Email	Employee’s email	
	Prenom	Employee’s first name	
	Role	Role of the Employee	

	Sexe	**Gender of the Employee**	
	Etat	**State of the Employee**	
	Date d'ajout	**Date of addition of the Employee**	
	Télephone	**Employee's phone number**	
Réunion	**ID Réunion**	**Identifier**	**Auto Incrément**
	Date	**The date and time of the meeting**	
	Lieu	**Meeting location**	
	Ordre du jour(desc)	**Report of the meeting**	
	Numpv reu	**Pv number of the meeting**	
	Nom projet	**Name of the associated project**	
	Heure	**Meeting time**	
Maitre ouvrage	**ID Mo**	**Identifier**	**Auto Incrément**
	Nom Mo	**Description of the manage service provider**	
	Type	**Type of the manage service provider**	
	Description	**Description of the manage service provider**	
	Adresse	**Address of the manage service provider**	
	Telephone	**Number phone of the manage service provider**	
	Email	**Email of the manage service provider**	
Maitre d'œuvre	**ID Mo**	**Identifier**	**Auton Incrément**
	Nom Mo	**Manage project owner's name (customer)**	
	Type	**Type of manage project owner**	
	Adresse	**Address of the manage service provider**	
	Telephone	**Number phone of the manage service provider**	
	Email	**Email of the manage service provider**	
	Description	**Description of the manage service provider**	
Marché	**ID mar**	**Identifier**	**Auto Incrément**
	Numero appel d'of	**Tender number**	

	Numero de contrat	Contract number	
	Prix dinar	Contract price in dinars	
	Type	Type of the contact	
	Date signature	Date of contract signature	
	Description	Description of the contract	
	Numéro appele dof mar	Contract tender number	
	Prix devise	Contract price in devises	
	Date visa cme	Date of visa cme	
	Visa cme		
Incident	ID incident	Identifier	Auto Incrément
	Description	Description of the incident	
	Date signalement	Date of incident reporting	
	Type incident	Type of the incident	
	Lieu	Location of the incident	
	L'heure	The time of the incident	
	Nom projet	Name of the associated project	
	Nom tache	Name of the associated task	
	Signalé par	The person who reported the incident	
tache	ID tache	Identifier	Auto Incrément
	IDchefprojet	Identifier	
	Nom tache	Task's name	
	Datedebut tache	The start date of the task	
	Datefin tache	The end date of the task	
	Statut	Status of the task	
	Nom projet	Name of the associated project	
	Description	Description of the task	
Suivi tache	ID	Identifier	Auto Incrément
	Date prevu	The expected start date of the task	
	Date réalisé	The date of completion of the task	
	Description	Description of the task tracking	

financier	ID financier	Identifier	Auto Incrément
	Nom	Financial 's name	
	Email	Financial's email	
	Prenom	Financial's first name	
	Role	Role of the Financial	
	Sexe	Gender of the Financial	
	Etat	State of the Financial	
	Date d'ajout	Date of addition of the Financial	
	Télephone	Financial's phone number	
Facture	ID facture	Identifier	Auton Incrément
	Numéro fac	Invoice description with number	
	ID mar	Identifier of contract	
	Fournisseur	Description of the supplier	
	Date fac	The date of the invoice	
	Date reception	The reception date of the invoice	
	Brut H.T	The tax rate	
	Montant Net H.T	The rate without amount H.T	
	Montant T.V.A	The rate without amount TVA	
	Montant T.T.C	The amount Net H.T plus TVA	
	Date Virement	Invoice transfer date	
	Ordre virement	Transfer ordre number	
	Désignation fac	Description of the invoice	
Fournisseur	ID utilisateur	Identifier	
	Nom	Name of the supplier	
	Prénom	First name of the supplier	
Suivie du projet	ID	Identifier	Auto Incrément
	Date prvu	The expected date to realize the project	
	Date réalisé	Date of completion of the project	

	Description	**Description of task tracking**	
	Type	**Type of project tracking**	

Figure 18:tableau de description des classes

Appendix 3

1. Implementation languages:

Python :

Figure 19:logo Python

Python is a multi-paradigm, versatile, interpreted, and high-level programming language. Python allows programmers to use different programming styles to create simple or complex programs, achieve faster results, and write code almost as if they were speaking in a human language. Some of the popular systems and applications that have used Python during development include Google Search, YouTube, BitTorrent, Google App Engine, Eve Online, Maya, and iRobot machines. (Python Software Foundation, 2025)

TypeScript :

Figure 20:logo TypeScript

TypeScript has a special relationship with JavaScript. TypeScript offers all the features of JavaScript, as well as an additional layer: its type system. (Microsoft Corporation, 2025)

HTML :

Figure 21:logo HTML

TypeScript has a special relationship with JavaScript. TypeScript offers all the features of JavaScript, as well as an additional layer: its typing system. HTML stands for "HyperText Markup Language," which can be translated as "markup language for hypertext." It is used to create and represent the content of a web page and its structure. Other technologies are used with HTML to describe the presentation of a page (CSS) and/or its interactive features. (JavaScript). (MDN Web Docs, 2025)

Css :

Figure 22:logo Css

CSS is one of the main languages of the open Web and has been standardized by the W3C. This standard evolves in the form of levels, CSS1 is now considered obsolete, CSS2.1 corresponds to the recommendation, and CSS3, which is divided into smaller modules, is in the process of being standardized. (MDN Web Docs, 2025)

2. The Tools Used::

Figma :

Figure 23:logo Figma

Figma combines powerful design tools with multi-user collaboration, allowing teams to explore their ideas while capturing quality feedback in real-time, at any moment.. (Figma, 2025)

Postman :

Figure 24:logo Postman

Postman is an all-in-one API platform for creating and using APIs. It simplifies every step of the API lifecycle: from design and testing to delivery and monitoring. Designed for teams, Postman simplifies collaboration, organization, and the rapid development of secure and reliable APIs. (Postman, 2025)

Git :

Figure 25:logo git

Git is a distributed version control system, free and open source, designed to manage all types of projects, from the smallest to the largest, with speed and efficiency.
Git is easy to use and offers minimal overhead, while providing ultra-fast performance. It surpasses version control tools like Subversion, CVS, Perforce, and ClearCase thanks to features such as economical local branch creation, convenient staging areas, and multiple workflows. . (Software Freedom Conservancy, 2025)

3. The interfaces :

3.1. the admin interface suite:

3.1.1 Add a user:

To add a new user, simply click on the "Créer nouveau" button to open a new page where you can enter the necessary information such as name, surname, gender, phone number, email address, and the user's role.

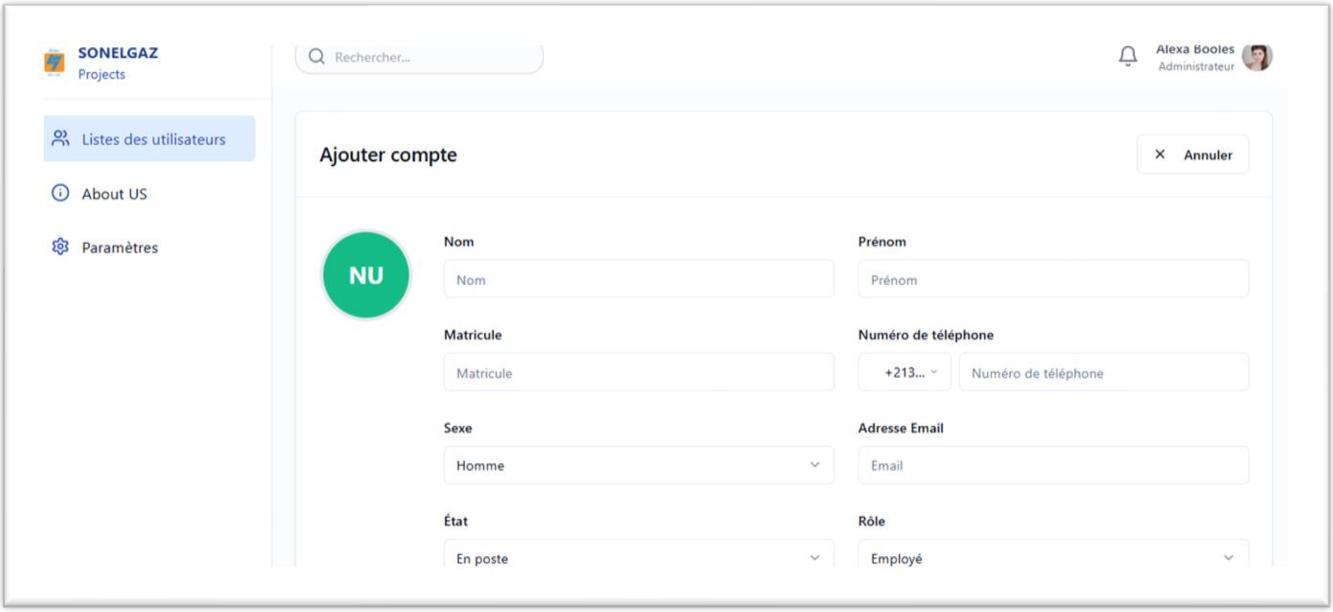


Figure 26:add user

3.1.2 Modify and delete a user:

If the admin wishes to modify an existing user account, they can change everything except the phone number and email address. Then, they press "Enregistrer" to record the changes in the database, or they can delete it by pressing "Supprimer."

3.2. Project interface :

3.2.1. Add project:

This interface represents the project addition form, and it is only accessible from the project manager's interface. To create a new project, the project manager will click the "Créer nouveau" button, which opens a form to be filled out. He must fill in the necessary information, then click on "Save" to add the project to the database. Once registered, the project becomes visible in the project manager's interface as well as in that of the concerned employees/financier, depending on the defined assignments.

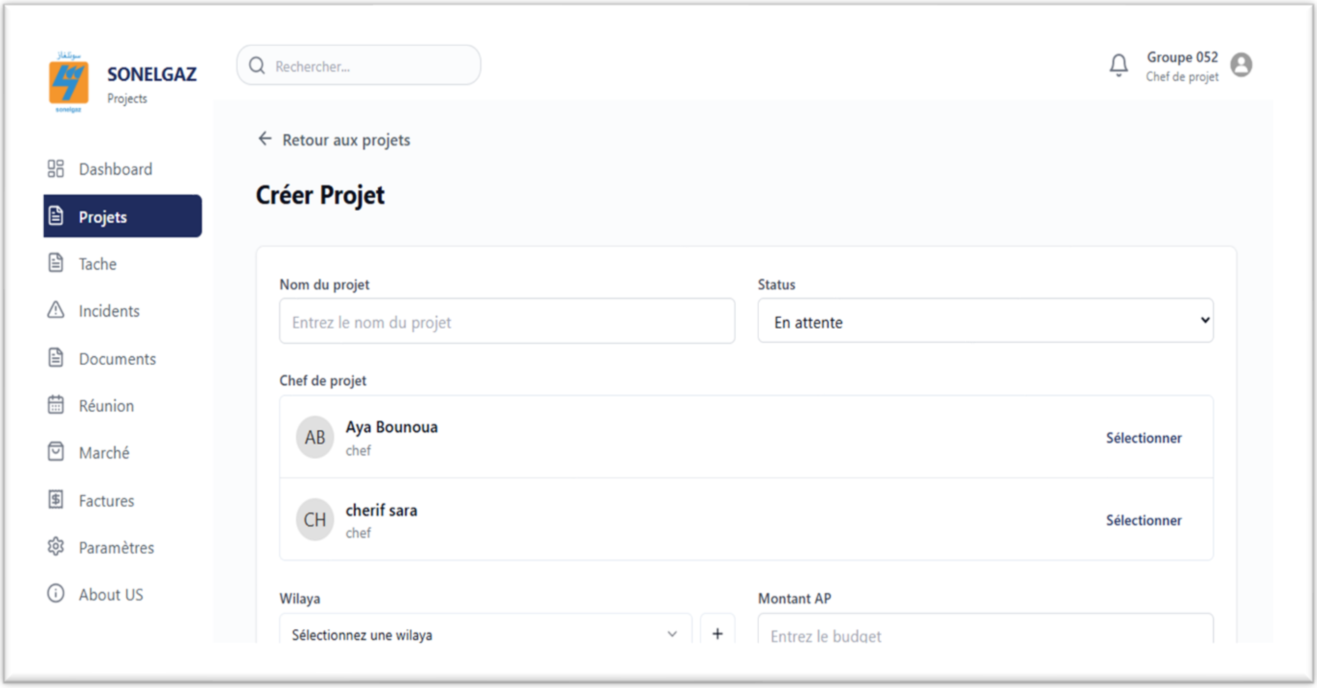
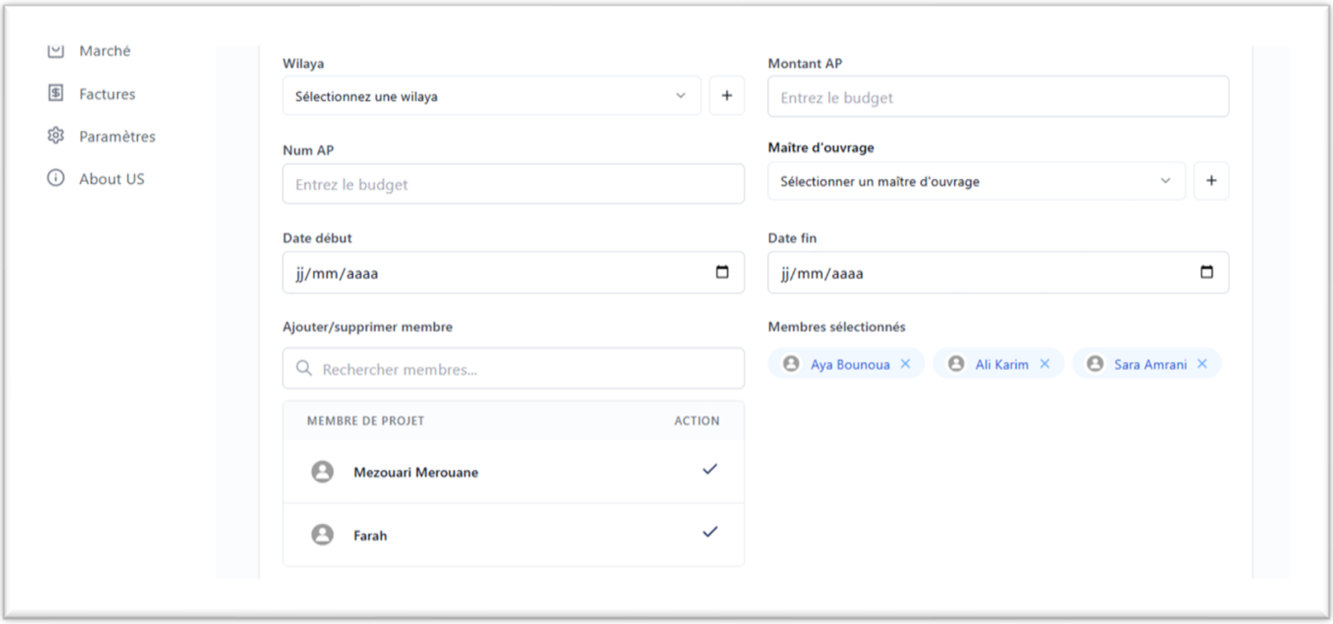
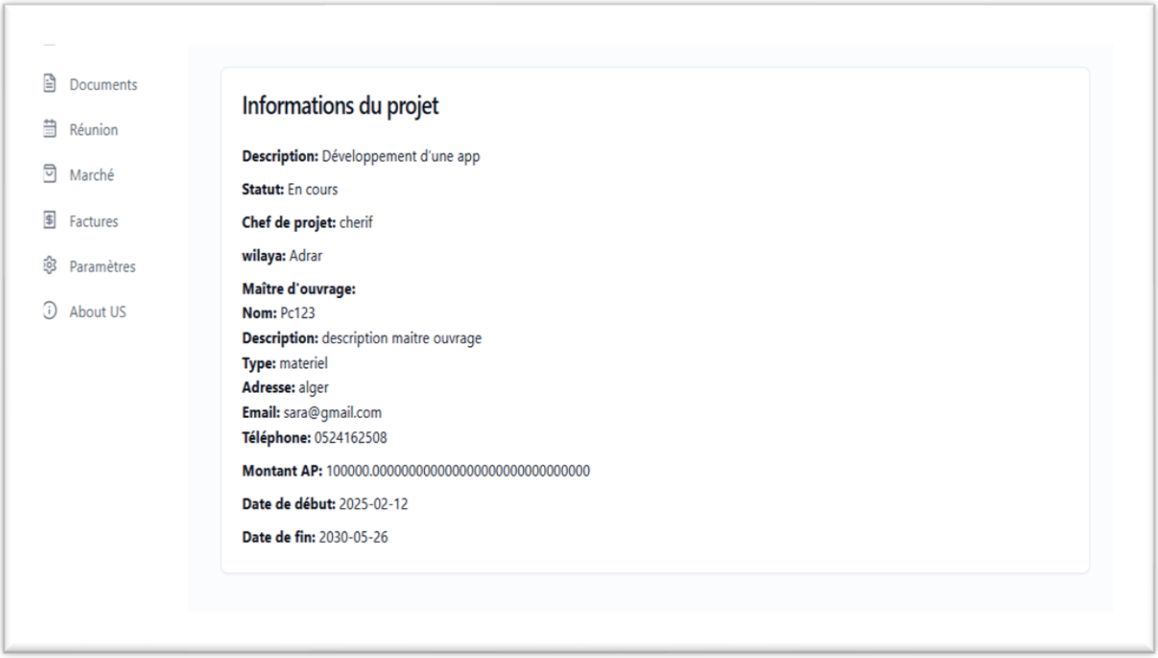
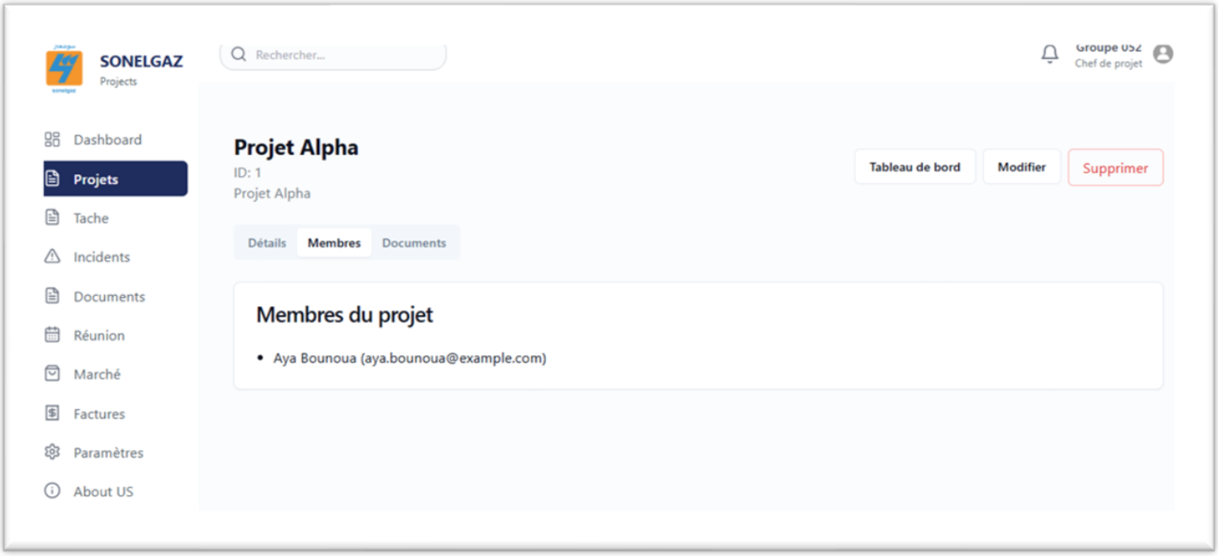


Figure 27:Add project



<u>Project details :</u>


Documents
Réunion
Marché
Factures
Paramètres
About US
Informations du projet
Description: Développement d'une app
Statut: En cours
Chef de projet: cherif
wilaya: Adrar
Maitre d'ouvrage:
Nom: Pc123
Description: description maitre ouvrage
Type: materiel
Adresse: alger
Email: sara@gmail.com
Téléphone: 0524162508
Montant AP: 100000.000000000000000000000000000000
Date de début: 2025-02-12
Date de fin: 2030-05-26


SONELGAZ
Projects
Rechercher...
Chef de projet
Dashboard
Projets
Tache
Incidents
Documents
Réunion
Marché
Factures
Paramètres
About US
Projet Alpha
ID: 1
Projet Alpha
Tableau de bord
Modifier
Supprimer
Détails
Membres
Documents
Membres du projet
Aya Bounoua (aya.bounoua@example.com)

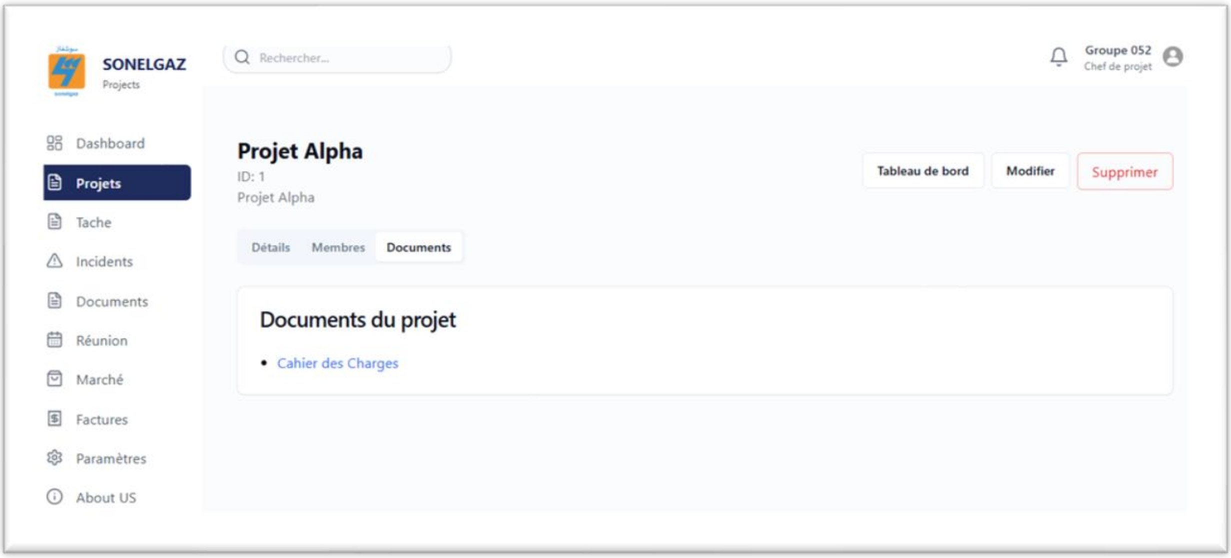


Figure 28:Project detail

3.3. task :

This interface displays the list of all tasks.
The "project manager" user can add, modify, delete, and view tasks only for their projects, assigning each task to the relevant employees.
The "employee" and "financial" users can view the tasks they are working on.

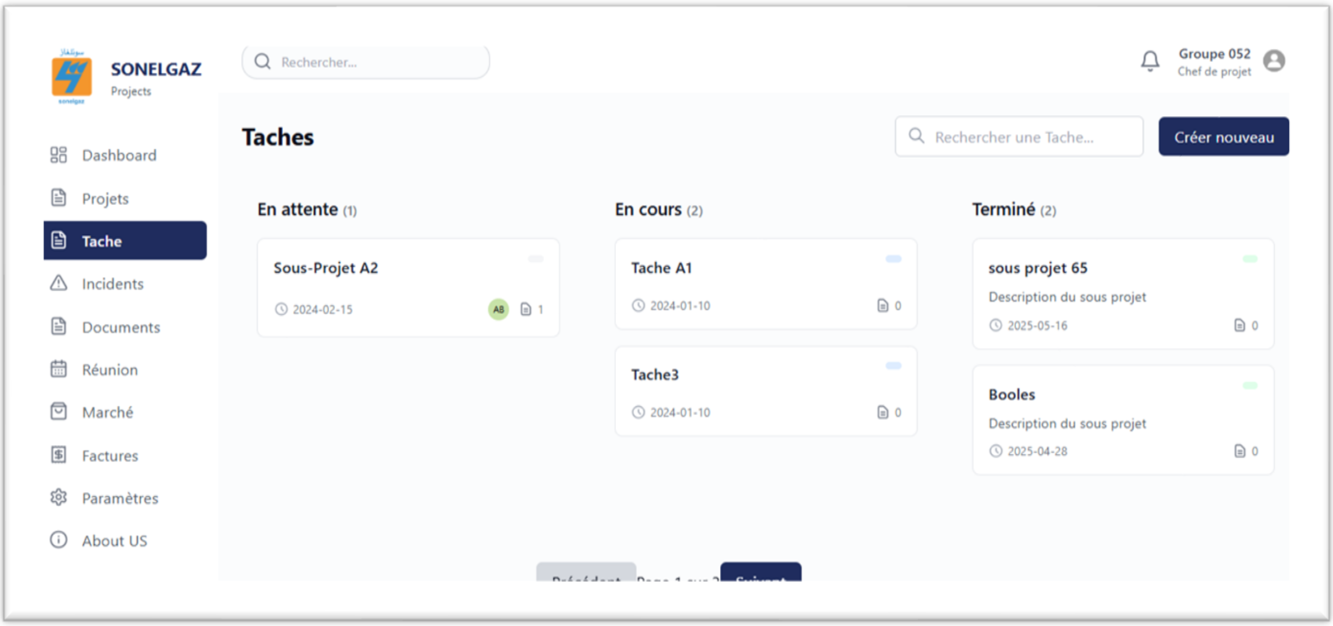


Figure 29: Tasks

3.3.1. Add task :

This interface represents the task addition form, and it is only accessible from the project manager's interface. To create a new task, the project manager will press the "Créer nouveau" button, which opens a form to be filled out. He must fill in the necessary information, then click on "Enregistrer" to add the task to the database. Once registered, the project becomes visible in the project manager's interface as well as in that of the concerned employees/financier, depending on the defined assignments.

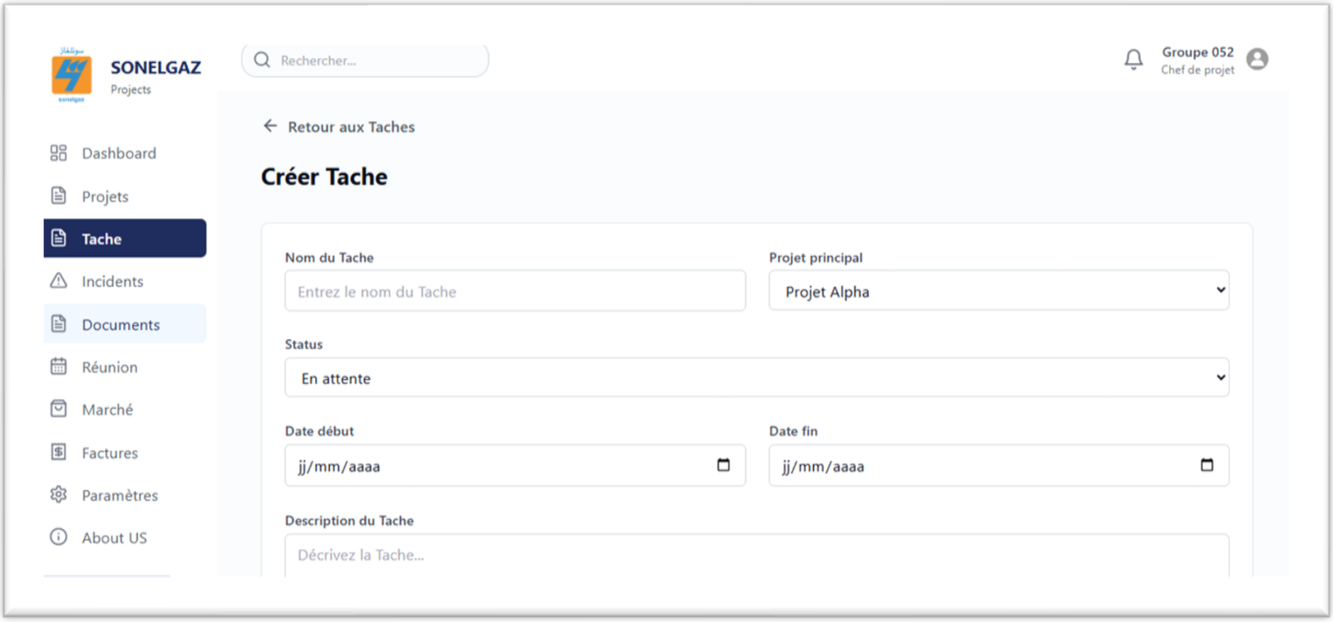
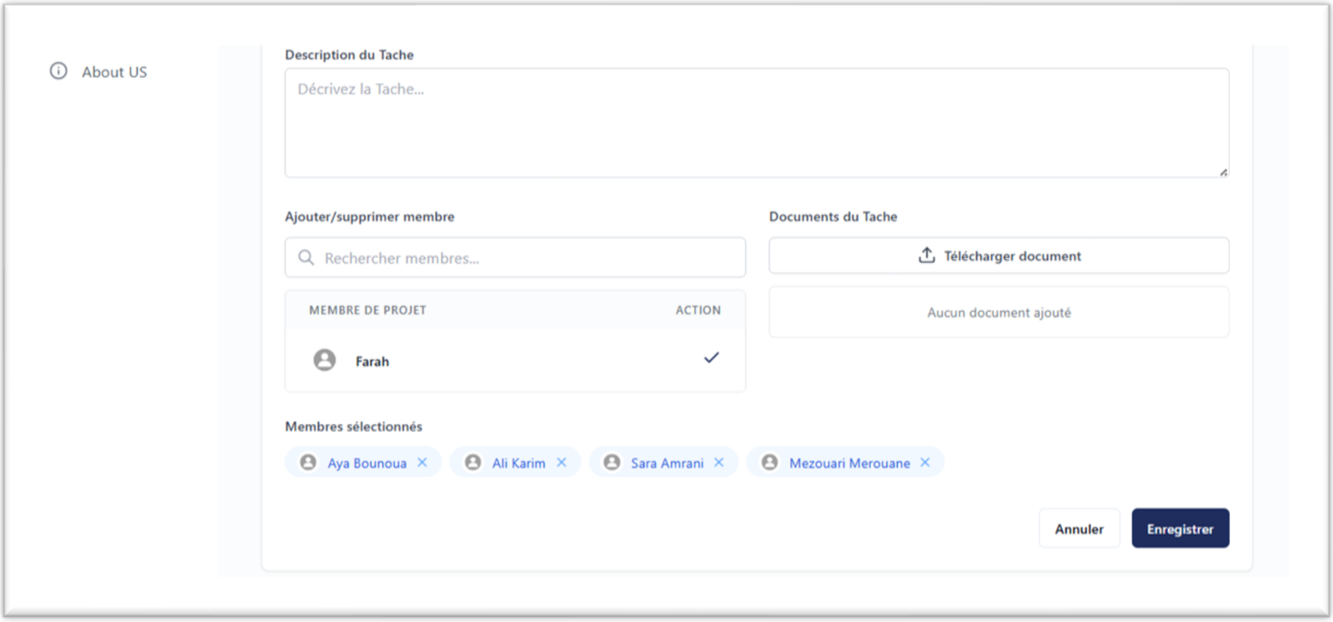


Figure 30 : Add task



3.3.2. Task detail :

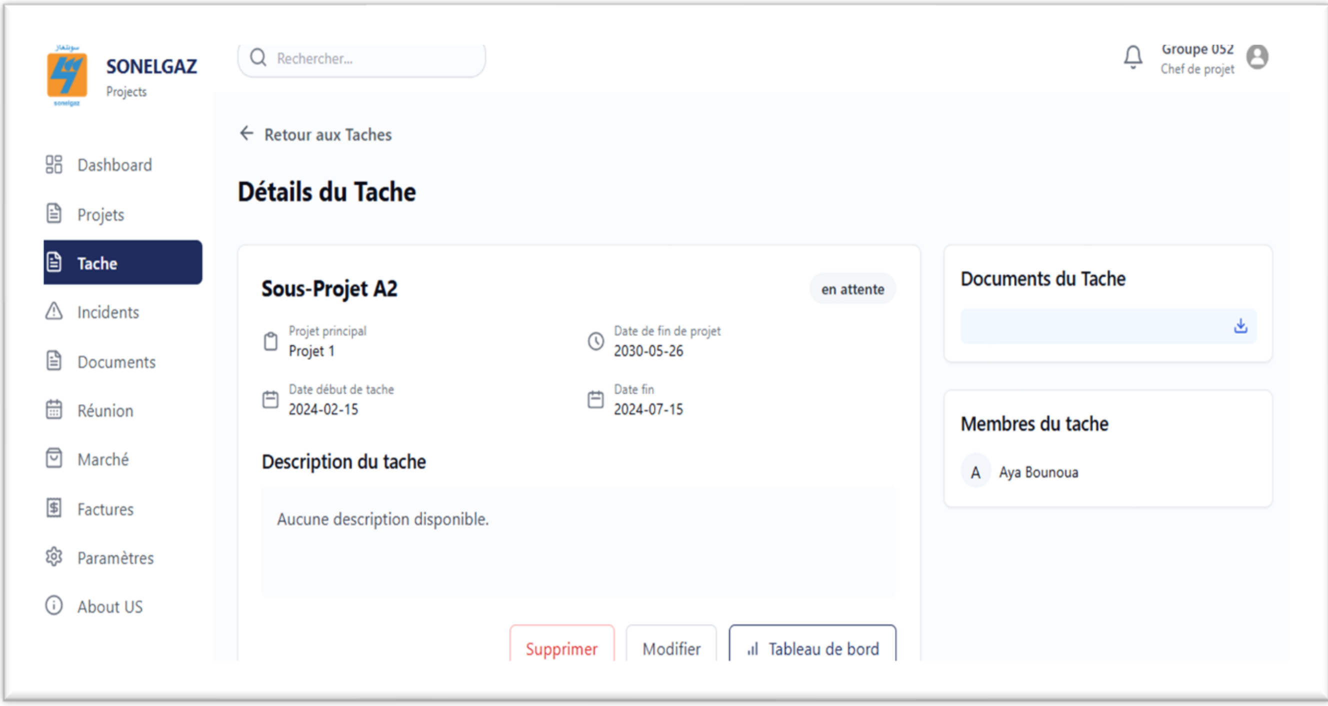
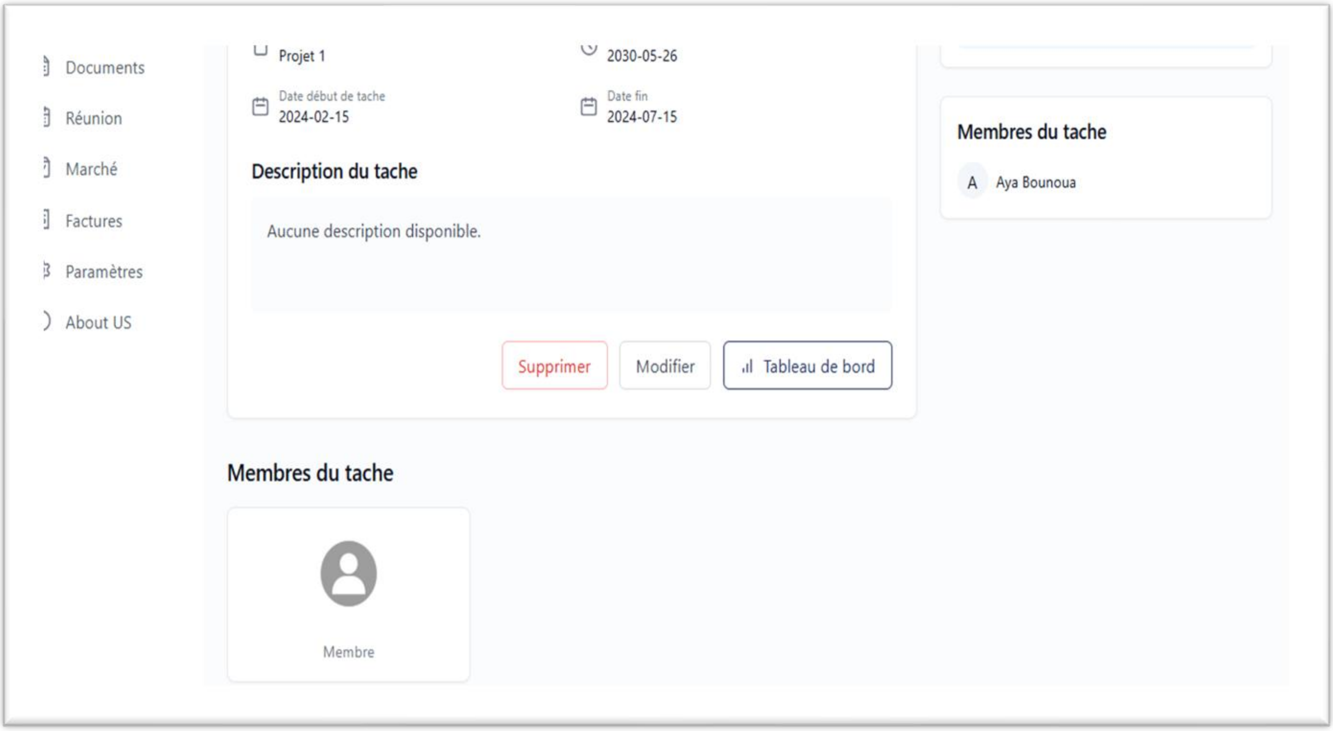


Figure 31:Task detail



3.4. Incidents :

This page is used to report and track all incidents during the execution of projects. All users except the admin access an interface that displays the list of incidents recorded for all projects carried out within Sonelgaz. From this interface, the project manager has several functionalities: View the details of a specific incident Edit incident information, Delete an incident, Add a new incident. this incident management ensuring better

traceability of events.
And for the other users, they can just consult it.

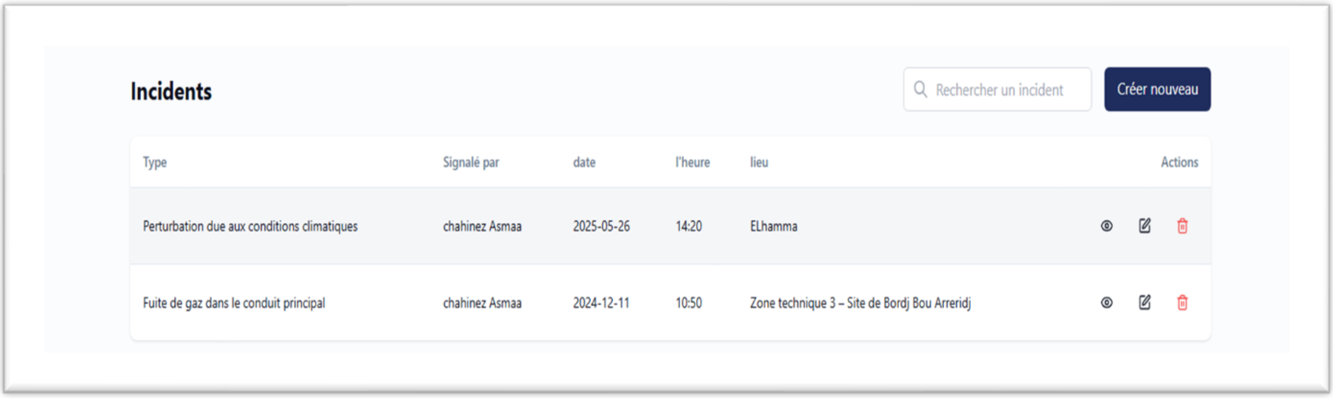


Figure 32:incident

3.4.1. <u>Add incident :</u>

For the manager to add a new incident, simply click the « Créer nouveau » button to open a new page where the necessary information can be entered.

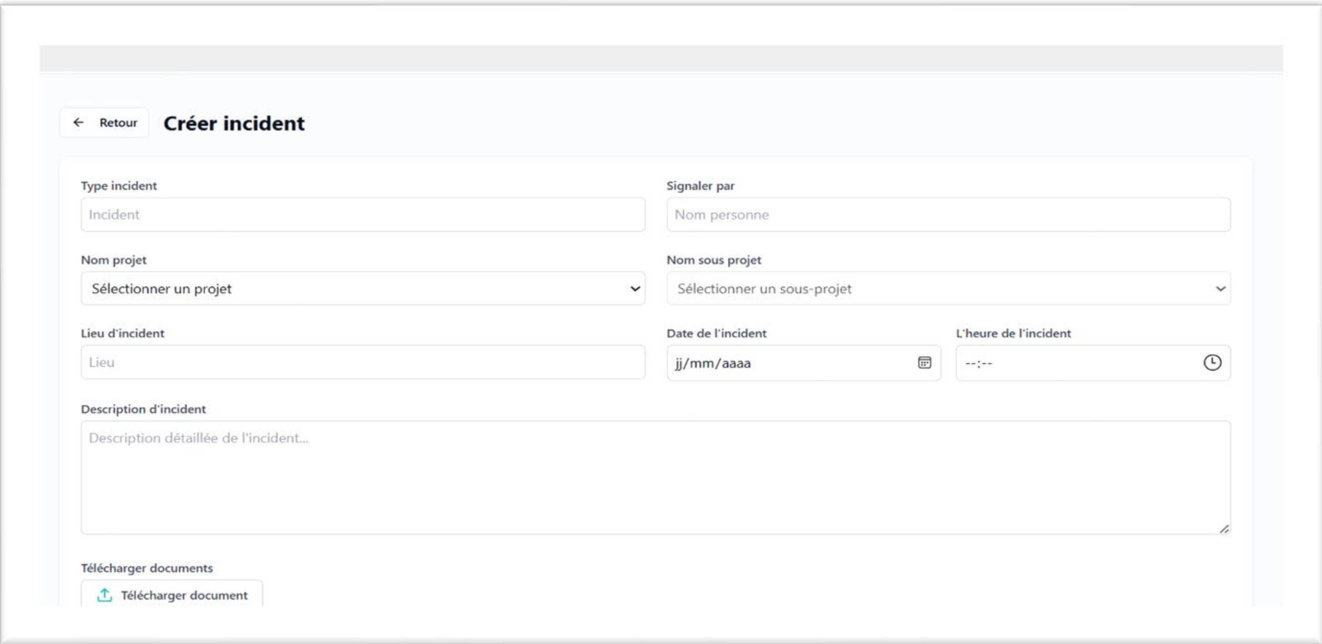


Figure 33: Add incident

3.4.2. <u>Incident detail:</u>

Figure 34:Incident detail

3.5. Document :

This interface displays the documents shared on the application with their information (details, projects, and tasks). It appears for both the project manager and the employees and financiers, who can then consult the document after sharing it. To view the details of a document, simply press the "eye" icon.

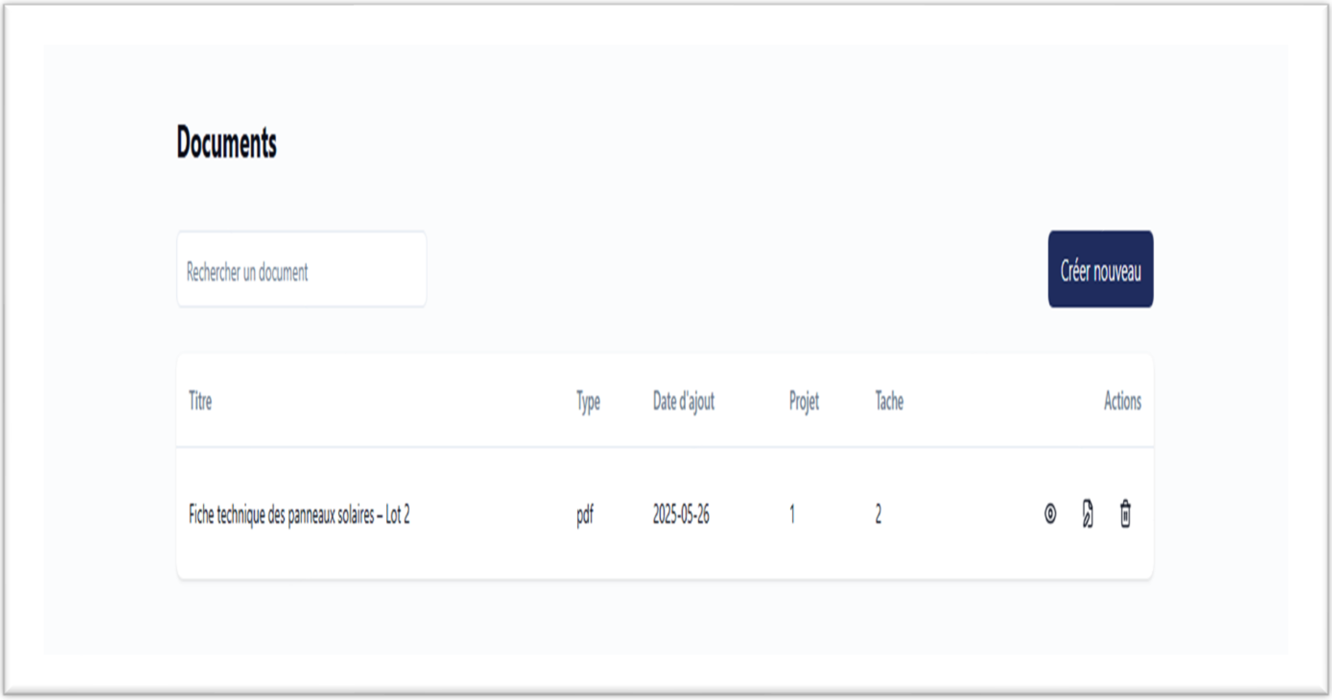


Figure 35:Document

3.5.1. Add document :

To add a new document, simply click on the bouton « Créer nouveau » button to open a new page where you can enter the necessary information.

.

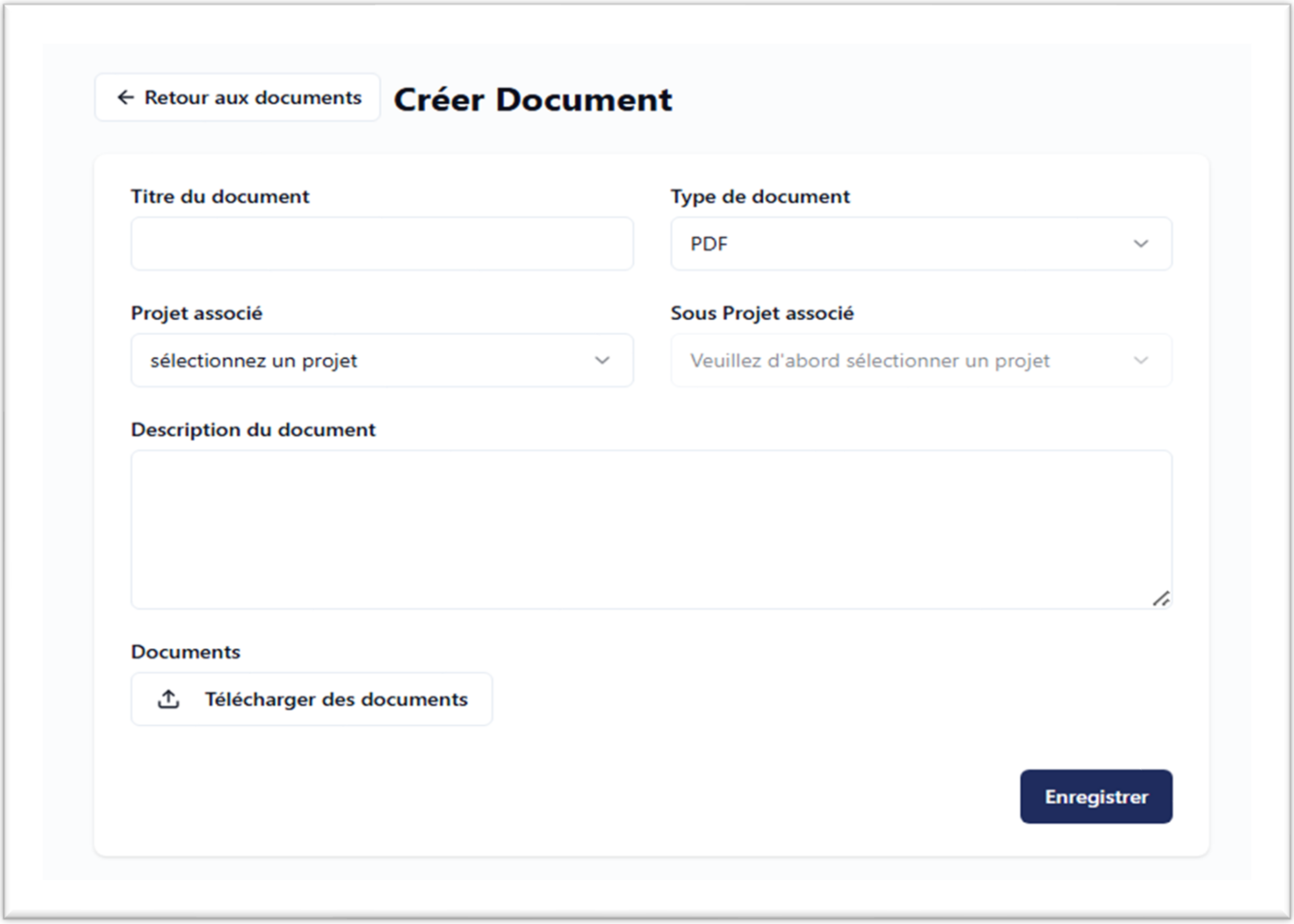


Figure 36:Add document

3.6. Meetings:

This page allows for the declaration and tracking of all meetings organized during the project execution. The project manager, the employee, and the financier access an interface displaying the list of recorded meetings. Except for the project manager who can add a new meeting, modify or delete a meeting, the other two users can only view the meetings.

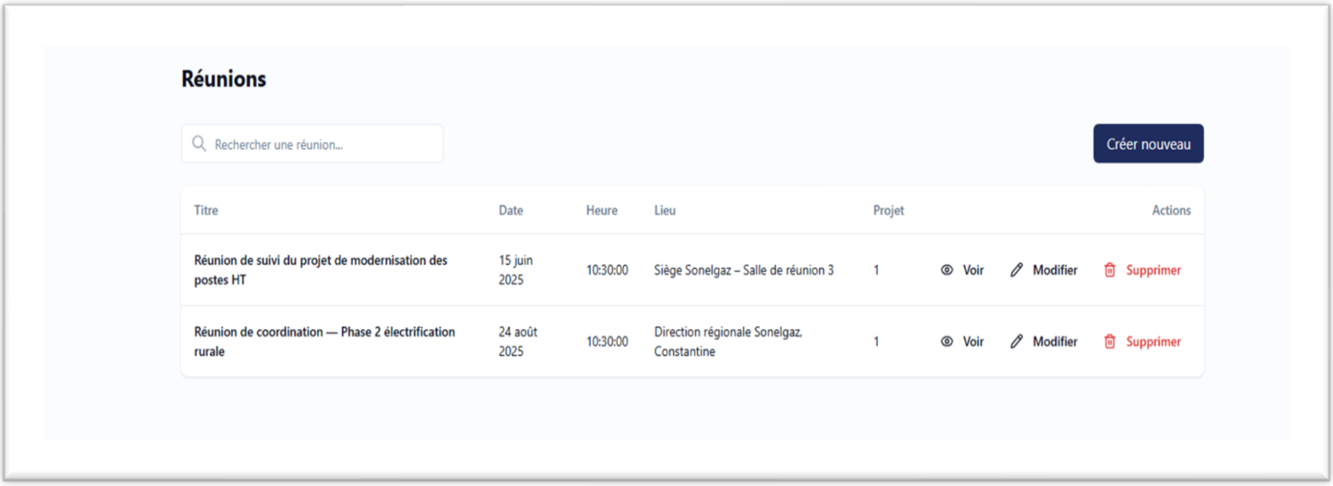


Figure 37:Meetings

3.6.1. Add a meeting :

To add a new meeting, simply click the « créer nouveau » button, which opens a form to enter the necessary information.

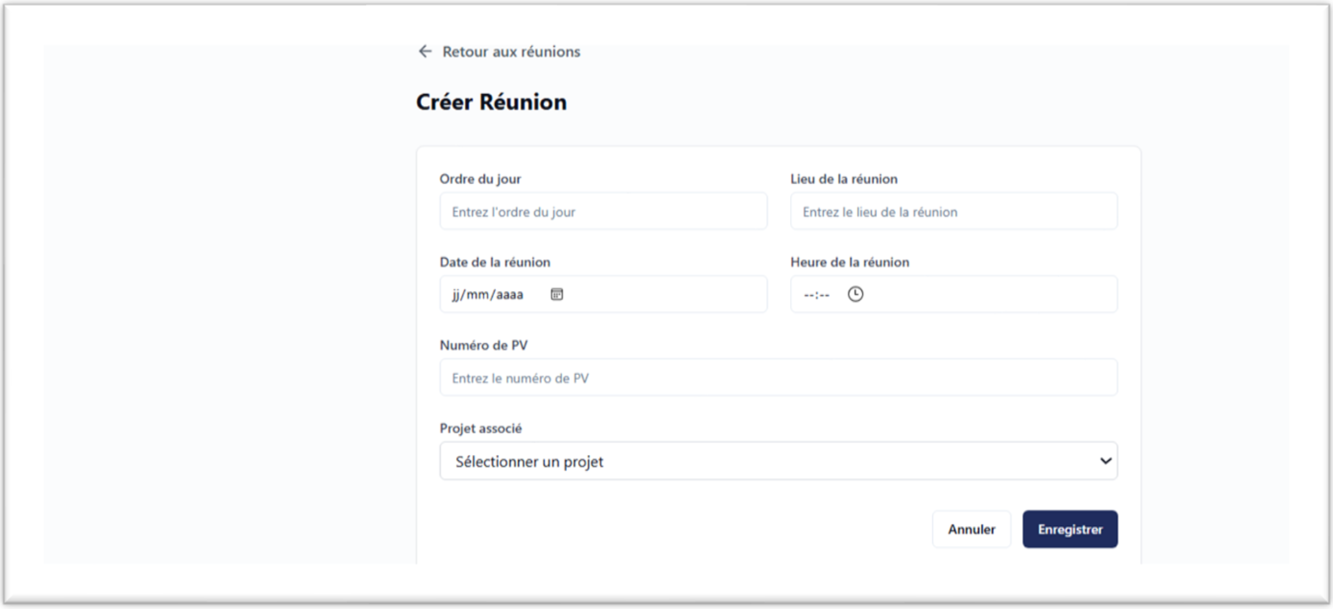


Figure 38:Add meeting

3.7. Market :

This interface shows the list of markets with their fields filled in, only the manager can handle this management. The employee and the financier can only consult it.

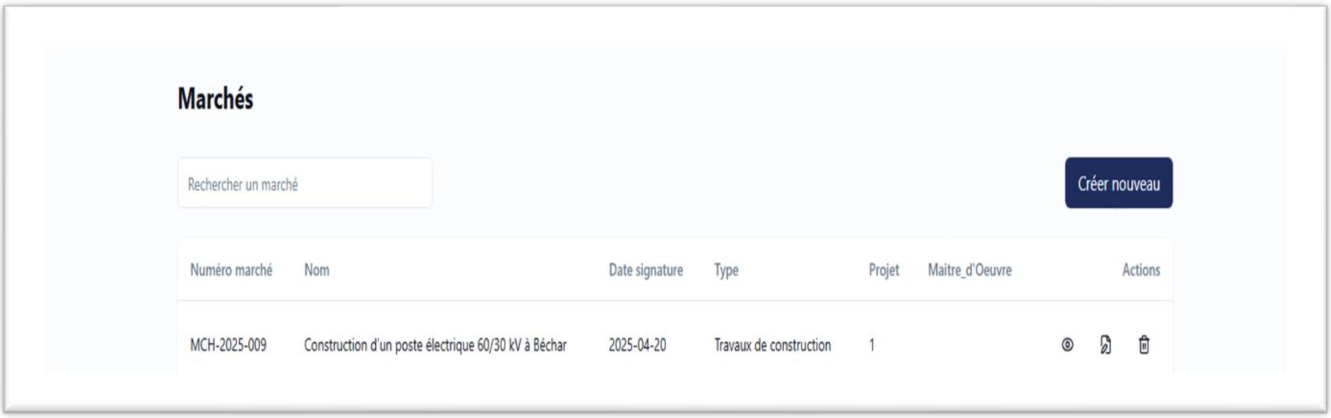


Figure 39:Market

3.7.1. Add market :

Pour ajouter une nouvelle réunion, il suffit de cliquer sur le bouton « créer nouveau »,button, which opens a form to enter the necessary information.

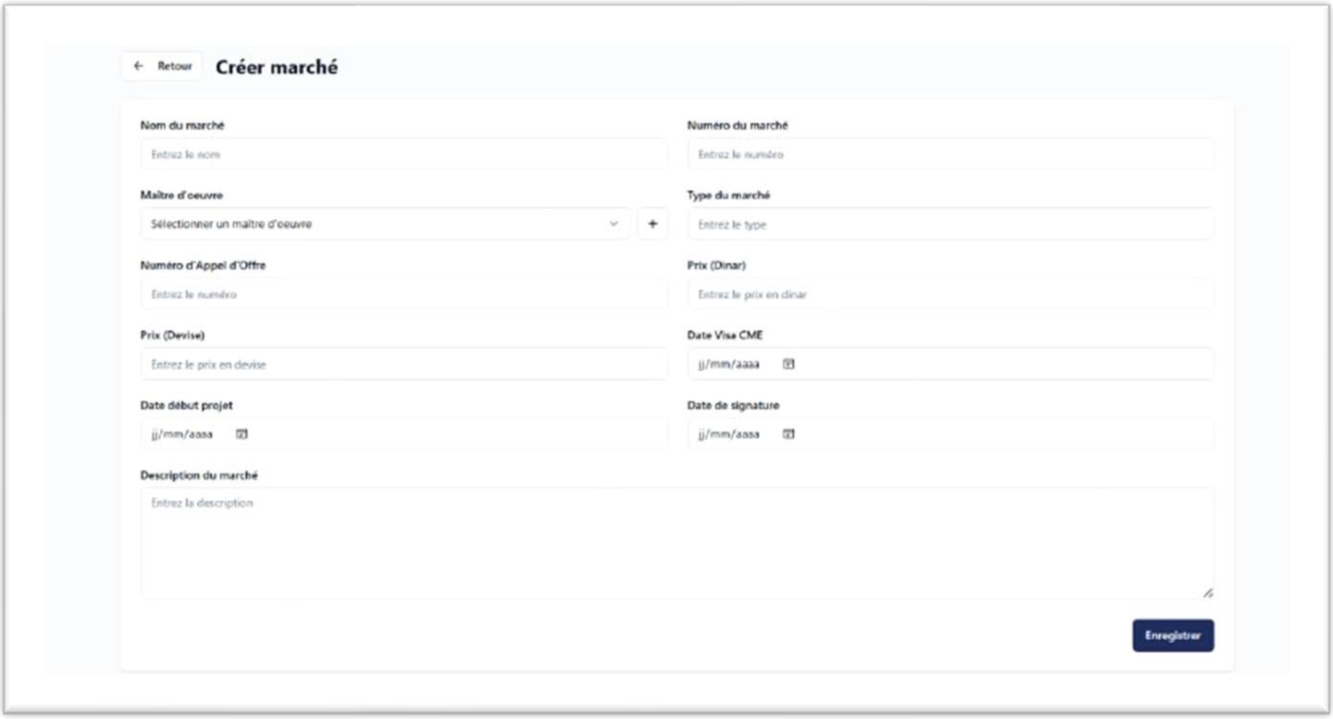


Figure 40:Add market

3.8. The invoices :

This interface displays the invoices shared on the application with their information. the project manager and the financial officer
The users "project manager" and "financial" can add, modify, delete, and view invoices only for their projects. The "employee" user can only view the invoices of their own projects.

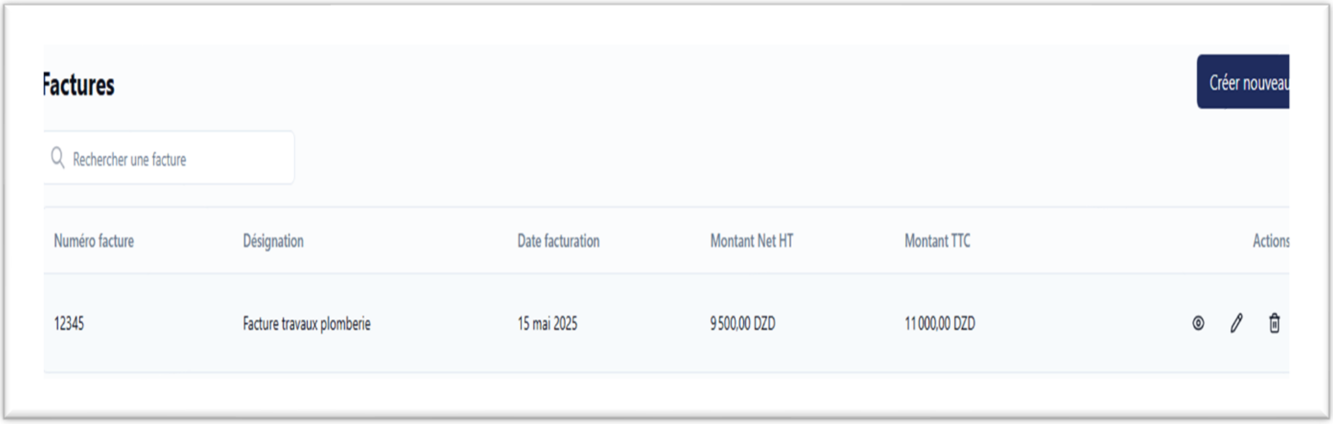


Figure 41: Invoice

3.8.1. Add invoice :

To add a new invoice, simply click on the « créer nouveau » button, which opens a form to enter the necessary information.

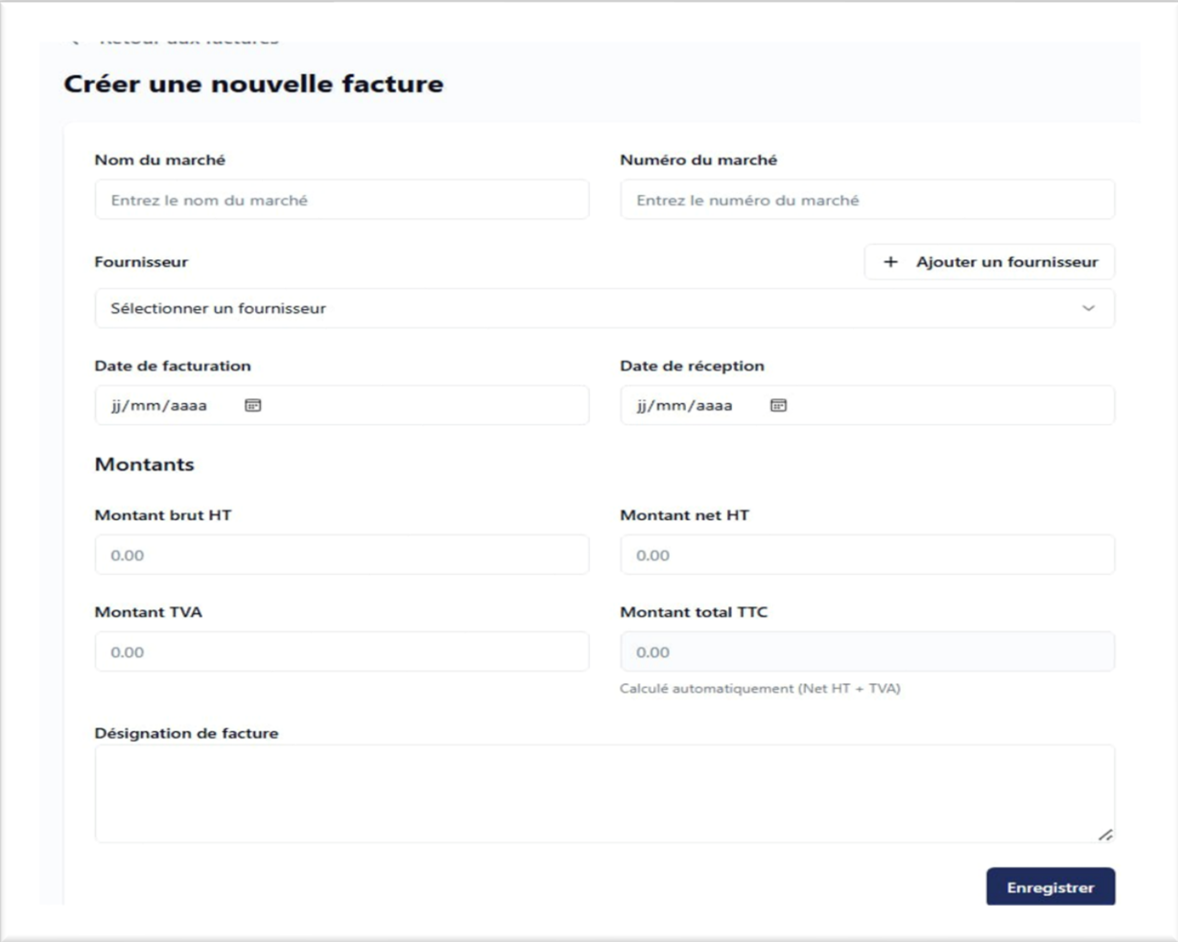


Figure 42:Add invoice

3.9. Profile:

This interface allows each user to view and manage the information of their own profile.
To access their personal information, the user can click on the "view my profile" button. He will then be redirected to a page displaying the details of his account, such as his first name, last name, email address, phone number, employee ID, and role.

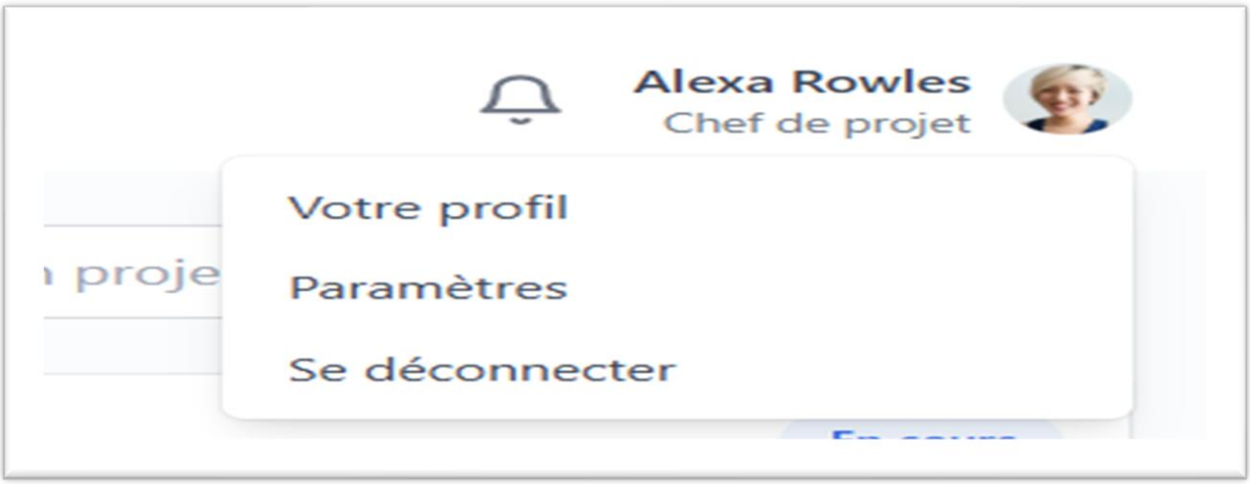


Figure 43:Profile

If the user wishes to update some of their information, they can click on the "Modifier profile" button. He will then have the option to modify his personal data, except for his matricule and role, which are permanently defined by the administrator to ensure the integrity of the system.

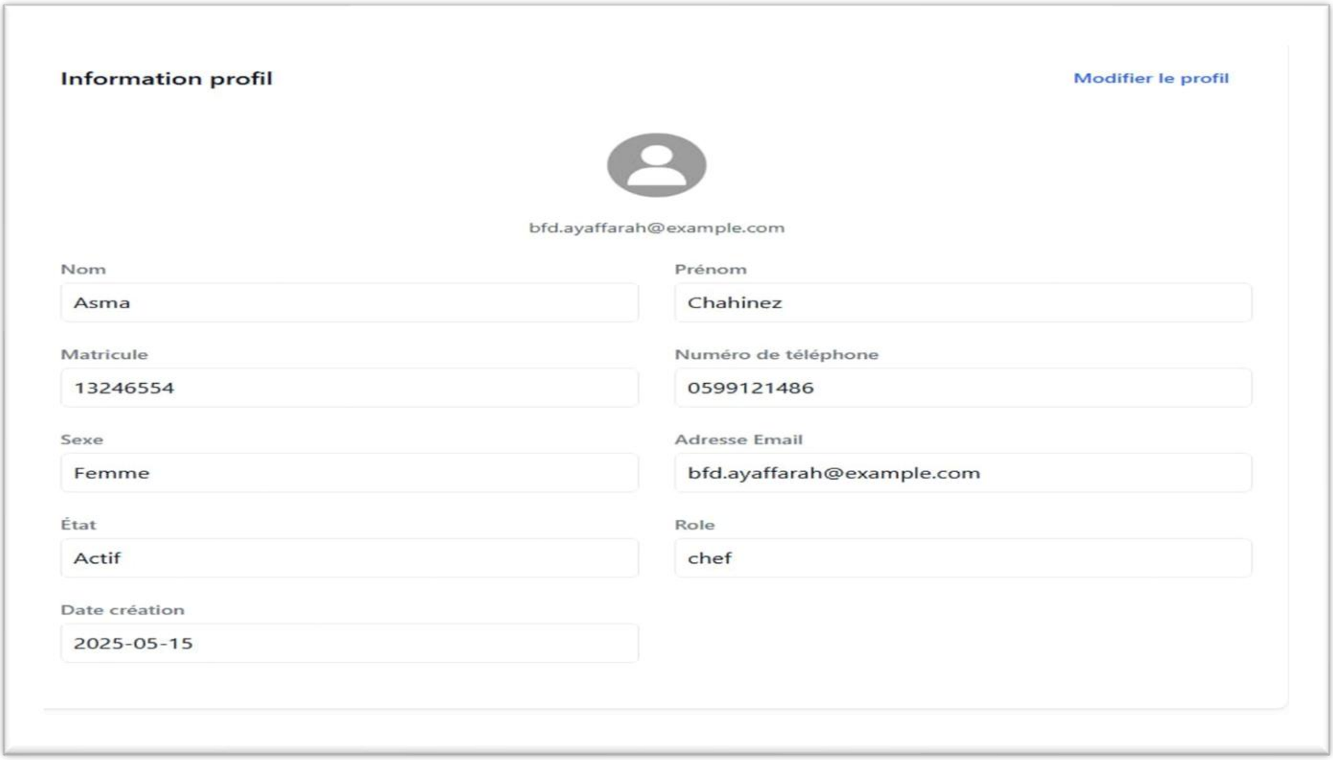


Figure 44:Profile informations

3.10. Intermediate tables

Certain specific entities are represented in the application in the form of intermediate tables, used for project and task management. These tables are accessible through dedicated interfaces in the management pages.

3.10.1. Project Owner (Customer) :

The Project Owner table represents the final client — that is, the entity for whom the project is being carried out. In the context of the application, **we are the project owner**, since our organization defines and manages the projects to be implemented.

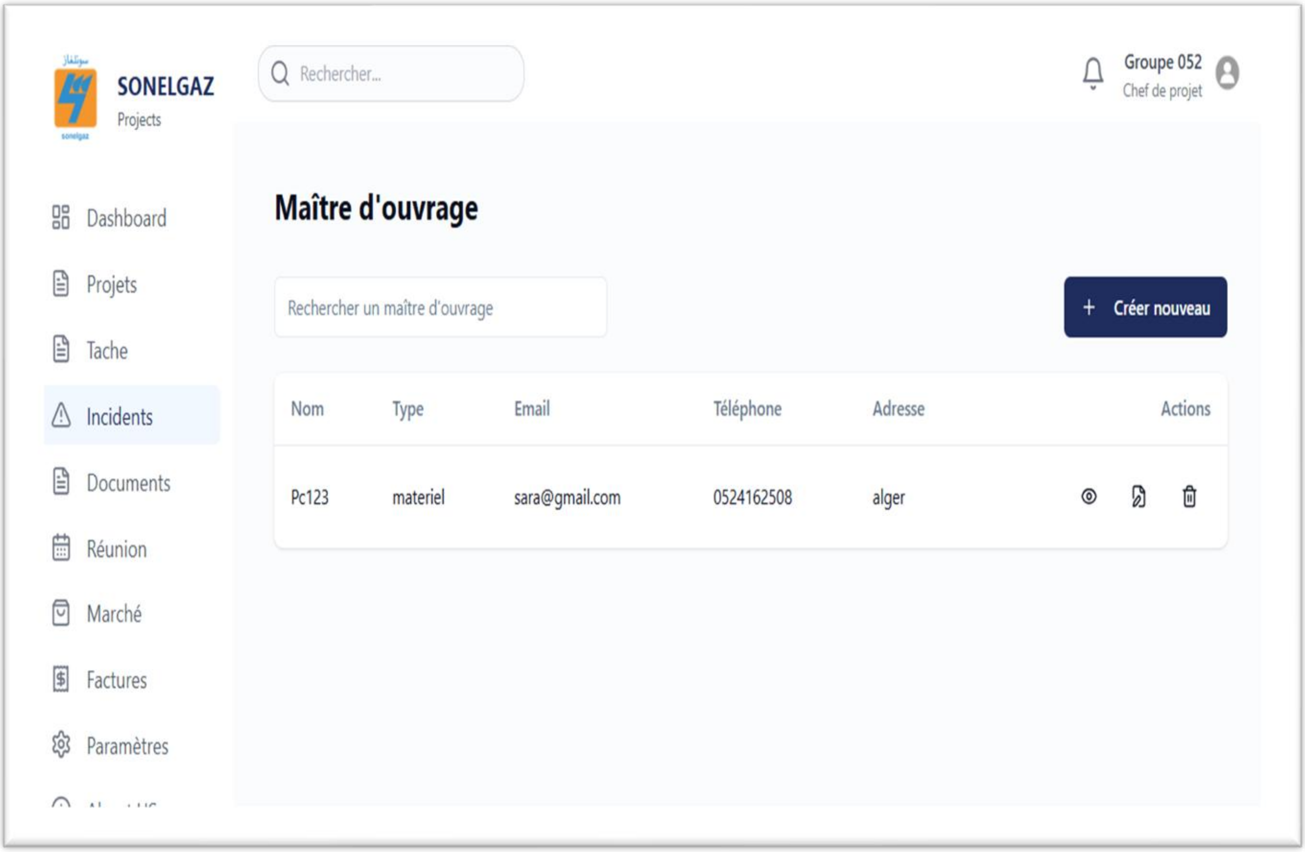
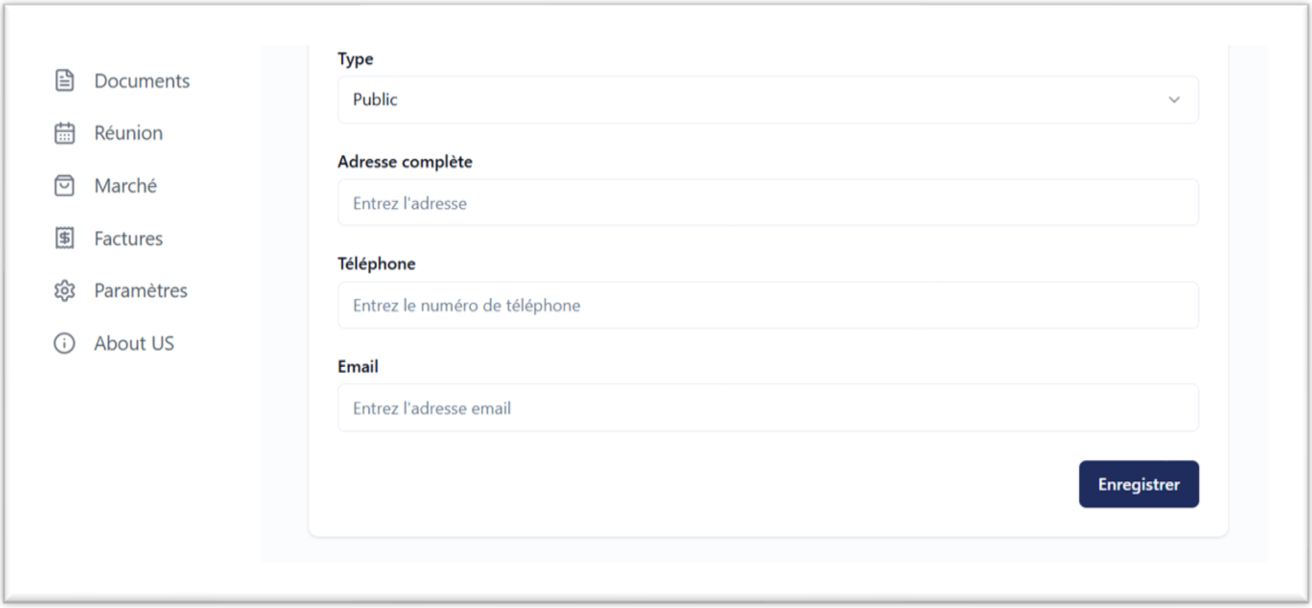


Figure 45:Project Owner (Customer)

Figure 46:Add Project Owner (Customer)



3.10.2. Service Provider:

In contrast, the Service Provider table represents the entity or company responsible for executing the project or part of the tasks. In this case, we act as the client, and the service provider is the external party in charge of delivering the work.

3.10.3. The suppliers :

The suppliers table groups external individuals or companies providing materials, services, or resources necessary for the successful execution of the project. Each record contains, in particular, the supplier's first and last name.

These three tables are integrated into the interface via management pages, allowing the authorized user to view, add, modify, or delete records as needed for the project.

3.11. Settings page :

This interface allows the user to change their password.

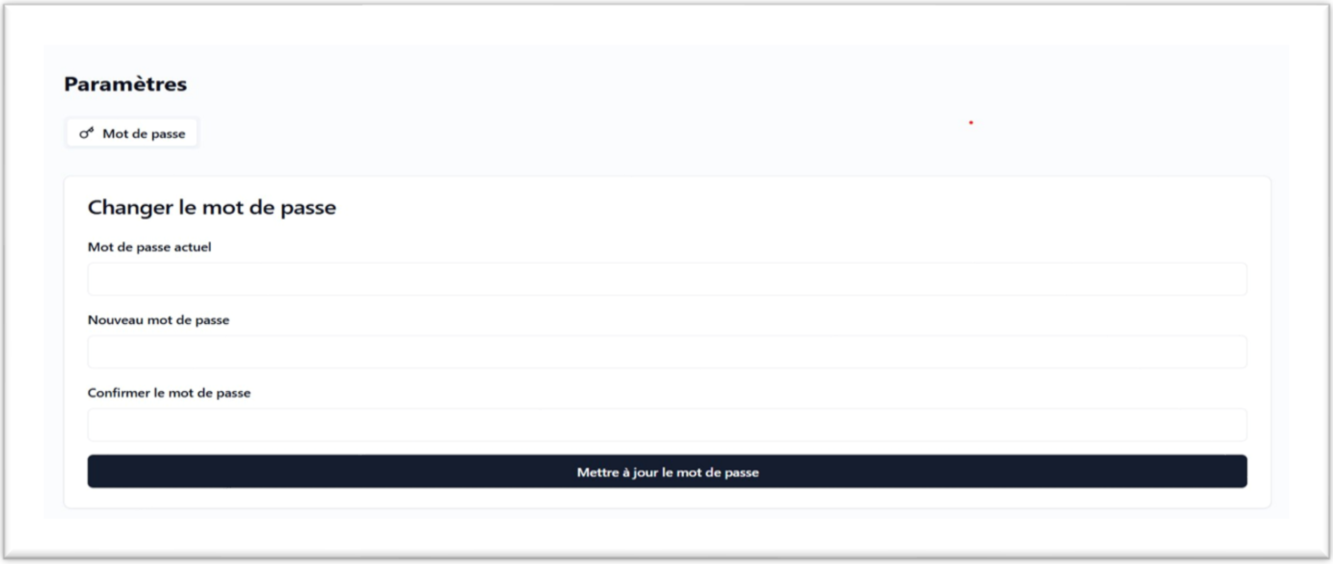


Figure 47:Settings